Heterogeneous Difference-in-Differences for Monadic Treatments in Dyadic Flow Data*

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Abstract: How do country-level policy changes affect bilateral flows with all partners? Estimating such monadic effects in dyadic data is challenging because (i) treatment is perfectly collinear with directional-time fixed effects used in applied work, (ii) heterogeneous pair-specific exposure to global shocks violates parallel trends, and (iii) network-wide equilibrium adjustments contaminate possible workarounds to replace directional-time fixed effects. We develop a common correlated effects difference-in-differences estimator that addresses all three issues. The approach augments pair-level PPML treatment regressions with cross-sectional averages of 'fundamental' variables constructed from never-treated countries, absorbing unobserved common shocks and equilibrium spillovers. Our core identifying assumption is testable using a diagnostic we introduce. Simulations calibrated to realistic trade structures show negligible bias and correct coverage. Using our method to study democratic regime changes, we find that they raise bilateral exports by 50% after two decades, with half of this effect operating through income changes.

Keywords: difference-in-differences, dyadic flow data, monadic treatments, heterogeneous treatment effects, common correlated effects, democratic regime change

JEL codes: C23, F13, F14, P16

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1 Introduction

The heterogeneous difference-in-differences (het-DID) revolution has transformed causal inference in panel data. Recent contributions have documented that the standard two-way fixed effects (TWFE) estimator can produce severely biased estimates when treatment effects vary across units or over time, particularly under staggered adoption [Goodman-Bacon, 2021, de Chaisemartin and D'Haultfœuille, 2020, Callaway and Sant'Anna, 2021, Sun and Abraham, 2021, Borusyak et al., 2024]. Robust alternatives now exist that avoid the negative weighting and forbidden comparison problems inherent in pooled TWFE estimation [Roth et al., 2023], including for nonlinear models [Wooldridge, 2023, 2025].

Many policy-relevant questions involve monadic treatments in dyadic flow data: a country-level intervention that reshapes the country's bilateral flows with all its partners. Does democratic regime change alter export patterns? Do capital controls redirect cross-border equity flows? Does visa liberalisation alter migration? Does regulatory harmonisation expand or divert trade in services? In each case, the treatment D_{it} varies at the country-time level while the outcome $Y_{ij,t}$ is observed at the country-pair-time level. The dyadic structure of the data is shared across these settings; the underlying economic model is not.

The workhorse empirical specification for bilateral flows includes directional-time fixed effects (μ_{it}, ν_{jt}) to absorb time-varying country-level unobservables. With a bilateral treatment such as a free trade agreement or a bilateral investment treaty, this works because the policy varies across pairs within the same origin-time cell. With a monadic treatment, D_{it} varies at exactly the same level as μ_{it} and is perfectly absorbed. The treatment effect is unidentifiable; this is a rank deficiency, not a power issue.

A natural alternative is to replace the directional-time fixed effects with explicitly constructed equilibrium adjustment terms used as continuous controls. This approach, however, faces two distinct problems. First, the adjustment terms are post-treatment objects: they are non-linear functions of every country's primitive characteristics, and a monadic treatment moves those characteristics through several channels at once. Conditioning on the adjustment terms therefore partials out the channels through which treatment operates, the textbook mediation/'bad control' problem. Second, in finite samples, the constructed proxies do not, in general, span the underlying common factor space, so residual factor confounding from heterogeneous pair-level exposure to global shocks remains in the residual. The two failure modes combine in a way that recovers neither the total reduced-form effect nor a clean partial effect. Absorbing common time effects cannot help either, since the network response to common shocks is *pair-specific* (depending on both countries' characteristics and on their position in the network), so a single μ_t absorbs only the average shift.

We develop the heterogeneous Common Correlated Effects Difference-in-Differences (CCE-DID) estimator to address these challenges. Cross-sectional averages of country-level variables computed from never-treated units sit upstream of the equilibrium (they are not mediators of treatment) and, provided they load on the relevant factors, span the common factor space (no residual confounding). Pair-specific coefficients on these averages allow each bilateral relationship its own sensitivity to global shocks, accommodating the heterogeneous exposure that network structure generates. Mean Group aggregation avoids negative weighting by construction. The key identifying assumption, factor spanning, is less restrictive than parallel trends and is directly testable through a diagnostic we develop, the 'CSD diff' test.

We make four contributions. First, we resolve the identification challenge inherent in *pooled* treatment models of monadic DID for dyadic data by augmenting *pair-wise* regressions with cross-sectional averages, thereby capturing confounding variation without absorbing the treatment. Second, we formalise the mediator-bias problem created by the network structure of dyadic flows, showing that proxies for equilibrium adjustment terms cannot serve as controls in our setup because they are post-treatment mediators, and demonstrate that fundamental-variable CAs from never-treated countries avoid this contamination. A large-scale simulation study calibrated to realistic bilateral trade structures establishes that this estimator is unbiased with correct coverage, while documenting substantial bias from alternative specifications, including omitting CAs, using all-country CAs, and controlling for equilibrium adjustment terms. Third, we develop a diagnostic procedure that directly assesses the factor spanning identification assumption with high power against economically meaningful violations and further propose a simple test for treatment bias from 'carryover' effects in samples with treatment reversal. Fourth, we apply our method to estimate the reduced-form treatment effect of democratic regime change on bilateral exports for a large set of countries over 1950–2014, finding that democratisation raises exports by approximately 50% after two decades, with half of the effect operating through the income channel.

Our work contributes to several literatures. Within the het-DID literature, we extend these methods to settings where the treatment unit (country) and the outcome unit (country-pair) do not coincide, and where network-wide equilibrium propagation creates complex contamination channels. Our factor model-based approach relates to Gobillon and Magnac [2016], Xu [2017], Chan and Kwok [2022], and Liu et al. [2024], who use interactive fixed effects to capture or impute counterfactual measures, but these methods have not been adapted to the dyadic setting with its additional challenges to identification. Furthermore, by delivering *pair-specific* treatment estimates, our method enables researchers to highlight patterns in the results that extend beyond standard ATET or dynamic treatment effect plots.

We build on the common factor panel literature [Pesaran, 2006, Boneva and Linton, 2017] by adapting the CCE approach to a causal inference setting in dyadic data, where the causal hierarchy of variables used for CA construction plays a central role. Chen et al. [2021] show that nonlinear factor models can capture important features of dyadic network data including degree heterogeneity, homophily in latent variables, and clustering, and confirm in a network application that multiple latent factors are empirically relevant. Whereas these authors estimate the full nonlinear factor structure, we use CCE proxies constructed from never-treated controls, thereby avoiding high-dimensional factor estimation and the associated bias corrections. We further study a distinct estimand, monadic treatment effects under staggered adoption in dyadic panels, and develop a heterogeneous DID estimator with pair-specific effects.

In the empirical literature using bilateral flow data, several recent contributions address related but distinct identification problems. Nagengast and Yotov [2025] apply het-DID methods to bilateral treatments, where treatment varies at the pair-level and is directly identified within standard structural approaches using directional-time fixed effects. Heid et al. [2021], Beverelli et al. [2024] and Freeman et al. [2025] develop structural approaches for country-specific policies in bilateral settings, imposing parametric assumptions and estimating average (pooled) partial effects. Our contribution sits alongside this work, addressing the case of *monadic* treatments whose mechanisms span multiple structural primitives, and where the total reduced-form effect is the policy-relevant target. For applications where a clean partial-effect decomposition is available, our framework recovers it as a special case (Section 3.2).

The remainder of the paper is organised as follows. Section 2 describes the setup. Section 3 presents the CCE-DID estimator and discusses inference. Section 4 summarises simulation evidence on finite-sample properties and introduces factor spanning and carryover diagnostics. Section 5 applies the estimator to study the effect of democratic regime change on bilateral exports. Section 6 concludes.

2 Setup

Consider bilateral flow data $Y_{ij,t}$, representing trade, capital, or migration flows from origin i to destination j at time t, where the treatment $D_{it} \in \{0, 1\}$ is a monadic (unit-level) binary indicator. Treatment is staggered (different units adopt at different times) and treatment may reverse. For ease of exposition, and in light of our empirical application, we consider democratic regime change in the setting of bilateral trade flows between countries when developing the empirical setup and in the simulation exercises in Section 4. Let $Y_{ij,t}(d)$ denote the potential outcome for pair (i, j) at time t when origin i's treatment status is $d \in \{0, 1\}$, with the observed outcome $Y_{ij,t} = D_{it} \cdot Y_{ij,t}(1) + (1 - D_{it}) \cdot Y_{ij,t}(0)$. For a total of N origin countries, we have N_{tr} treated and N_c never-treated (control) countries.

A subtlety arises from the network structure of dyadic flows. Standard potential outcomes assume that one unit's treatment does not affect another unit's outcome (SUTVA). In dyadic flow data this is violated through a specific mechanism: treatment changes country i's primitive characteristics, these enter a system-wide equilibrium as inputs, and the equilibrium solution shifts for all countries, altering flows even between untreated countries.

Our estimand, the *total reduced-form treatment effect*, is defined conditional on this equilibrium response. The potential outcome $Y_{ij,t}(1)$ denotes the flow observed when $D_{it} = 1$, given the full vector of treatment assignments realised in the data. The estimand is specific to the realised treatment history: it measures the effect of country i's treatment in the equilibrium that actually prevails, which depends on which other countries are treated simultaneously. It is the change in i's own flows, inclusive of whatever equilibrium adjustments occur, and it is the relevant object for ex-post policy evaluation when the treatment moves several structural primitives jointly. For applications where the treatment maps cleanly into a single structural parameter (for instance, a bilateral cost-shifter), a partial-effect decomposition is feasible and existing structural techniques are appropriate — Section 3.3 makes the relationship explicit.

In line with common practice in the literature on dyadic flow data, we model the outcome as an exponential linear model, but introduce a multi-factor error structure:

$$Y_{ij,t} = \exp\big(\alpha_{ij} + \boldsymbol{\lambda}'_{ij}\mathbf{F}_t + \theta_{ij} \cdot D_{it} + \delta_{ij} \cdot D_{jt} + \boldsymbol{\beta}'_{ij}\mathbf{X}_{ij,t}\big) \cdot \eta_{ij,t} \tag{1}$$

where F_t is a $K \times 1$ vector of unobserved common factors (capturing global business cycles, technology, commodity prices, financial conditions), λ_{ij} is a $K \times 1$ vector of pair-specific factor loadings, α_{ij} is a pair-specific fixed effect (capturing time-invariant bilateral characteristics), θ_{ij} is the pair-specific origin treatment effect (the object of interest), δ_{ij} captures the effect of destination treatment, $X_{ij,t}$ are additional dyadic controls with pairwise coefficients β_{ij}, and $\varepsilon_{ij,t}$ is an idiosyncratic error with $\mathbb{E}[\eta_{ij,t} \mid F_t, D_{it}, D_{jt}, X_{ij,t}] = 1$.

This formulation treats the origin as the primary treated unit but does not restrict destinations. Including D_{jt} absorbs destination treatment variation that would otherwise confound the origin effect θ_{ij}; our event-study analysis (Section 5.4) further exploits destination heterogeneity by stratifying estimates by destination regime status.

We further consider country-level variables that capture observable characteristics of each country, drawn from any source that loads on the common factor structure. In our empirical application we use country-level income and population, but the framework accommodates demographic shares, capital stock measures, electricity consumption, and other aggregates. Let $z_{m,it}$ denote the m-th such variable:

$$\ln z_{m,it} = \alpha_i^{z_m} + \gamma_i^{z_m\prime} F_t + \theta^{z_m} D_{it} + u_{m,it}, \qquad m = 1, \dots, M. \tag{2}$$

Some of these variables may respond to treatment ($\theta^{z_m} \neq 0$), others may not. The framework requires only that some such variables load on the common factor structure and remain available in the never-treated subsample, where θ^{z_m} is zero by construction.

Bilateral frictions (due to transaction costs, regulatory distance, informational barriers) also admit a factor structure:

$$\ln \tau_{ij,t} = \alpha_{ij}^{\tau} + \gamma_i^{\tau\prime} F_t + \gamma_j^{\tau\prime} F_t + \theta^{\text{cost}} D_{it} + \xi_{ij,t}. \tag{3}$$

As is standard in the factor-model panel literature [Pesaran, 2006, Bai, 2009, Chen et al., 2021, Chan and Kwok, 2022] we assume that the country-level variables and the outcome share a common factor structure. Under the maintained assumption of time-constant effects, the object of interest is the pair-level treatment effect $\theta_{ij} = \ln Y_{ij,t}(1) - \ln Y_{ij,t}(0)$ for all t. Channel decomposition (for instance, isolating the income channel) can proceed through mediation analysis (see Section 3.2), but the total effect is the natural starting point. An event-study analysis in Section 5.4 relaxes time-constancy to allow $\theta_{ij,\ell}$ indexed by event time ℓ.

3 Estimation and Inference

3.1 Construction of the CCE-DID Estimator

Cross-section averages. Our approach augments each pair-level regression with cross-sectional averages (CAs) that absorb confounding common-factor variation without absorbing the treatment. Following Pesaran [2006], consider the cross-sectional average of an observable $z_{c,t}$ but over never-treated countries $c \in \mathcal{C}$:

$$\bar{z}_t^{\text{ctrl}} = \frac{1}{N_c} \sum_{c \in \mathcal{C}} z_{c,t} = \bar{\alpha}_{\text{ctrl}} + \bar{\gamma}'_{\text{ctrl}} \mathbf{F}_t + \bar{u}_t^{\text{ctrl}} \tag{4}$$

where $\bar{\gamma}_{\text{ctrl}} = N_c^{-1} \sum_{c \in \mathcal{C}} \gamma_c$ is the average factor loading across control countries. As $N_c \to \infty$, $\bar{u}_t^{\text{ctrl}} \xrightarrow{p} 0$ by the law of large numbers, yielding:

$$\bar{z}_t^{\text{ctrl}} \approx \bar{\alpha}_{\text{ctrl}} + \bar{\gamma}'_{\text{ctrl}} \mathbf{F}_t \tag{5}$$

Since $\bar{\gamma}_{\text{ctrl}} \neq \mathbf{0}$, the control CA is a (rotated) proxy for the common factors. Including it as a regressor absorbs the common factor variation that would otherwise confound the DID comparison. More generally, with M distinct CA variables (each computed from a different observable), the CAs can span a factor space of dimension $K \leq M$. In our application, two CAs (GDP and population) yield $M = 2$ per country dimension, providing sufficient spanning capacity for a two-factor structure.

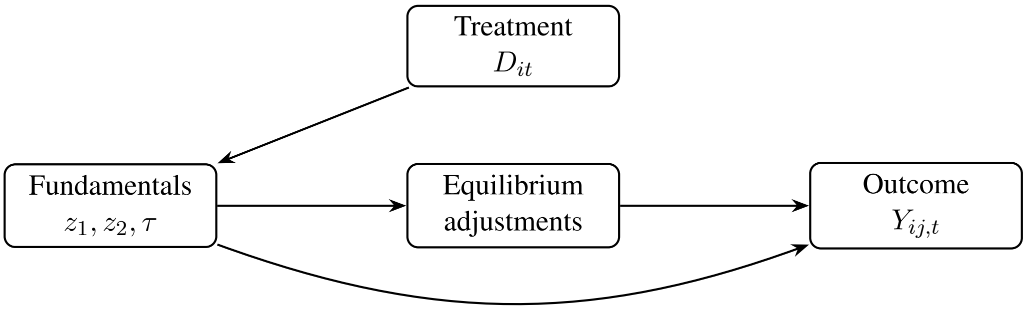
Two restrictions govern our CA construction. First, CAs must be computed from *never-treated* countries only. If all-sample CAs were used instead:

$$\bar{z}_t^{\text{all}} = \bar{\alpha} + \bar{\gamma}' \mathbf{F}_t + \theta_z \cdot \underbrace{\frac{N_{\text{tr}}}{N} \bar{D}_t}_{\text{treatment contamination}} + \bar{u}_t, \tag{6}$$

where $\bar{D}_t = N_{\text{tr}}^{-1} \sum_{i \in \text{treated}} D_{it}$ is the average treatment intensity among treated countries at time t and N_{tr}/N is the treated share of the sample. The all-sample CA absorbs part of the treatment effect, making it a bad control: a post-treatment variable that is itself a function of treatment. Control-only CAs avoid this contamination. Since control countries are never treated, no treatment term enters equation (4), and the cross-sectional average remains a clean factor proxy. This is the CCE analogue of the 'never-treated' restriction in the het-DID literature [Callaway and Sant'Anna, 2021, Chan and Kwok, 2022] and reflects the same principle as the 'forbidden comparison' concern raised by de Chaisemartin and D'Haultfœuille [2020]: treated units' outcomes must not enter the counterfactual.

The second restriction concerns which variables to average. One might want to use CAs of bilateral flows themselves or of proxies for the network-wide equilibrium adjustment terms, since these also load on $\mathbf{F}_t$. The causal hierarchy of the DGP (Figure 1) shows why this fails.

Figure 1: Causal hierarchy of the data-generating process



Notes: Treatment affects fundamentals (income, population, bilateral costs), which enter the system-wide equilibrium and determine bilateral flows both directly and through equilibrium adjustment terms. CAs should be constructed from variables as far upstream as possible.

Fundamentals (incomes, populations, bilateral costs) are model inputs [Head and Mayer, 2014]: they are determined by country-specific intercepts, factor loadings, treatment status, and idiosyncratic shocks *before* the equilibrium is solved.[1] For control countries ($D_{ct} = 0$), no treatment term enters their fundamental equations, so they reflect only factors and noise:

$$z_{m,ct} = \alpha_c^{zm} + \boldsymbol{\gamma}_c^{zm\prime}\mathbf{F}_t + u_{m,ct}, \qquad m = 1, \ldots, M \tag{7}$$

and their cross-sectional averages are clean factor proxies. The key insight is that the causal arrow runs one way: fundamentals feed into the equilibrium, but the equilibrium does not feed back into fundamentals. When country i is treated, the equilibrium adjustment terms of *every* country shift (including those never treated), but control countries' fundamentals are unaffected because fundamentals are determined *upstream* of the equilibrium adjustment.

[1] In the structural gravity model, income is an exogenous endowment. In practice, measured GDP includes trade-related value added, so control-country GDP could respond to treatment elsewhere through trade multipliers. This feedback is doubly attenuated in our setup: each treated country's demand shift is spread across up to $N-1$ partners, and exports are a fraction of each partner's GDP. Additionally, the CA terms are averaged over N_c control countries, further attenuating by $1/N_c$, hence the total order of attenuation can be $O(1/((N-1)\cdot N_c))$, which is second-order relative to the $O_p(N_c^{-1/2})$ CA approximation error. By contrast, MRT contamination is first-order: MRTs are defined as functions of *all* countries' fundamentals. Our simulations confirm that fundamental-based CAs produce negligible bias whereas specifications using MRTs are severely biased.

CAs of equilibrium objects (equilibrium adjustment terms, bilateral flows) inherit treatment contamination even when computed from control countries only, because these objects are solved from the full system that includes treated countries' changed fundamentals — we elaborate on the equilibrium-adjustment case in Section 3.3 and on the bilateral-flows case in Section 5.3. The guiding principle is therefore to construct CAs from variables that sit upstream of the equilibrium: control-country fundamentals satisfy this requirement by construction.

Estimation. The pair-level estimation equation is then:

$$\mathbb{E}[Y_{ij,t} \mid D_{it}, D_{jt}, \bar{\mathbf{z}}_t^{\text{ctrl}}] = \exp\big(\alpha_{ij} + \theta_{ij} \cdot D_{it} + \delta_{ij} \cdot D_{jt} + \boldsymbol{\phi}_{ij}' \bar{\mathbf{z}}_t^{\text{ctrl}}\big) \,. \tag{8}$$

Here α_{ij} is a pair fixed effect absorbing time-invariant bilateral heterogeneity, D_{it} is the origin treatment indicator, D_{jt} is the destination treatment indicator, $\bar{\mathbf{z}}_t^{\text{ctrl}}$ is the vector of CAs from never-treated countries, and $\boldsymbol{\phi}_{ij}$ is a vector of *pair-specific* CA coefficients. Including D_{jt} ensures that θ_{ij} captures the origin treatment effect net of any destination regime change occurring over the same period.

Dyadic controls (e.g., FTA membership, common currency) may be included in $\mathbf{X}_{ij,t}$:

$$\mathbb{E}[Y_{ij,t} \mid \cdot] = \exp\big(\alpha_{ij} + \theta_{ij} \cdot D_{it} + \delta_{ij} \cdot D_{jt} + \mathbf{X}_{ij,t}' \boldsymbol{\beta}_{ij} + \boldsymbol{\phi}_{ij}' \bar{\mathbf{z}}_t^{\text{ctrl}}\big) \,. \tag{9}$$

The PPML estimator is consistent under correct specification of the conditional mean, without requiring correct specification of the conditional variance [Santos Silva and Tenreyro, 2006]. The exponential form naturally accommodates zero flows and is robust to heteroskedasticity. The CCE consistency results of Pesaran [2006] are established for the linear case. Boneva and Linton [2017] extend the CCE to the setup with a binary dependent variable, which explicitly excludes the CA for the dependent variable. In an unpublished extension to their Probit-CCE they provide suggestions that the cross-section average augmentation developed is also valid for the generalised linear model, including a Poisson Maximum Likelihood implementation. Formal asymptotic theory for CCE-DID in the nonlinear setting remains to be developed, though in Appendix A we discuss sufficient conditions and sketches of the proofs. Our simulation study, which uses the PPML estimator throughout, confirms that the approach delivers correct bias and coverage properties in finite samples calibrated to realistic trade data.

The pair-specific coefficients ϕ_{ij} on the CAs reflect the structure of dyadic flow data. In network settings, pair-level factor loadings inherit dependence from both endpoints and from each country's position in the network of flows. Chen et al. [2021] formalise this for nonlinear factor models in network and panel data: the latent factor structure underlying dyadic outcomes generates pair-specific loadings that vary across the $N(N-1)$ directed pairs even when the underlying factor space has low dimension, capturing features such as degree heterogeneity, homophily in latent traits, and clustering. The relevant implication for our purposes is that the pair-level loading $\boldsymbol{\lambda}_{ij}$ on common factors $\boldsymbol{F}_t$ is a function of country-level loadings together with a network-position term that itself depends on the joint loading distribution. Whatever the precise functional form, pair-level sensitivity to common shocks is heterogeneous in a way that a common time effect cannot absorb.

This heterogeneity has three implications. First, the parallel trends assumption fails because treated pairs have systematically different λ_{ij} from control pairs. Second, pooled estimation fails because a common coefficient ϕ on the CAs cannot absorb pair-specific factor exposure. Third, Mean Group estimation is therefore necessary: estimating $(\alpha_{ij}, \theta_{ij}, \phi_{ij})$ separately for each pair and averaging allows each pair to absorb its own factor exposure. Pesaran [2006] showed in the two-way panel that the MG estimator is consistent for the average effect under loading heterogeneity, as long as a law of large numbers holds. Chan and Kwok [2022], specifying estimated factor proxies following Bai [2009] rather than CAs, extend this result to the DID setting and establish consistency under $N_{\text{tr}}, T \to \infty$ with $\sqrt{N_{\text{tr}}}/T \to 0$.

Robust Mean Group Aggregation and Inference. To aggregate the pair-specific $\hat{\theta}_{ij}$ into an overall/sample ATET, we use two complementary approaches. The first applies robust regression (iteratively reweighted least squares with Tukey's biweight function; rreg) to all pair-wise estimates:

$$\hat{\theta}_{ij} = \mu + e_{ij}, \qquad \hat{\theta}^{\text{rreg}} = \hat{\mu}, \tag{10}$$

where $\hat{\mu}$ is the robust intercept estimate, downweighting extreme outlier pairs.[2] The second is a two-step procedure that first computes the median $\hat{\theta}_{ij}$ per treated origin i:

$$\bar{\theta}_i = \text{median}\big(\{\hat{\theta}_{ij}\}_j\big), \qquad i = 1, \ldots, G \tag{11}$$

[2]We use the term ATET (*average* treatement effect) for the rreg aggregation, implying it targets the mean; however, this is only correct if the random coefficient assumption in Appendix equation (31) holds. More generally, rreg targets the median-centred location of pair-level effects, which under symmetry of e_{ij} coincides with the mean treatment effect.

and then applies rreg across the G origin-level medians. This is more robust to pair-level outliers because the within-origin median already removes extreme destinations before the across-origin aggregation step. Since we estimate the granular *pair-wise* treatment effect, researchers are free to aggregate treatment effects by alternative sub-groups, enabling them to highlight structural differences in the result (e.g., advanced vs. developing country effects) or to investigate 'mechanical' distortions (e.g., quantifying the bias from treatment reversal).

Inference must account for the dependence structure inherent in dyadic data. Pairs sharing an origin share both D_{it} and likely factor loading dependence, inducing within-origin correlation; pairs sharing a destination share importer-side general-equilibrium adjustments. Following Cameron et al. [2011], we use two-way clustering by exporter and importer, the standard approach for bilateral data: $\text{Var}_{\text{CGM}} = \text{Var}_{\text{exporter}} + \text{Var}_{\text{importer}} - \text{Var}_{\text{pair}}$, with $t(\min(G_{ex}, G_{im}) - 1)$ critical values, where $\text{Var}_{\text{exporter}}$ clusters by origin, $\text{Var}_{\text{importer}}$ clusters by destination, and Var_{pair} is the pair-level (robust) variance subtracted to avoid double-counting. The use of $t(\min(G_{ex}, G_{im}) - 1)$ critical values corrects for the smaller cluster dimension, following the recommendation for few-cluster settings. This procedure achieves correct coverage across simulation configurations (see Table B-4 in Appendix B).[3]

3.2 Identification

Factor Spanning. Our core identification assumption replaces parallel trends in conventional het-DID with a weaker factor spanning condition.

Assumption 1 (Factor Spanning). *Let* $\bar{\gamma}^{z_m}_{ctrl} = N_c^{-1} \sum_{c \in \mathcal{C}} \gamma^{z_m}_c$ *denote the average* $K \times 1$ *factor loading vector of the* m*-th CA observable across never-treated countries. Stack these into the* $M \times K$ *matrix* $\bar{\mathbf{\Gamma}} = [\bar{\gamma}^{z_1}_{ctrl}, \ldots, \bar{\gamma}^{z_M}_{ctrl}]'$. *The factor spanning condition requires: rank*$(\bar{\mathbf{\Gamma}}) = K$, *where* M *is the number of distinct CA variables and* K *is the number of common factors.*

This requires that the CAs from never-treated countries collectively capture all K factors driving outcomes, demanding $M \geq K$ with non-collinear average loading vectors. Unlike parallel trends, which requires homogeneous factor loadings, factor spanning permits each pair its own sensitivity to the common factors, identifying the treatment effect from residual variation after removing pair-specific factor exposure.

[3] In factor-augmented models, common factors induce residual cross-cluster dependence that no standard clustering method fully captures. CGM two-way clustering provides the best available approximation, dominating both one-way exporter clustering and ad-hoc inflation factors across all simulation configurations.

Factor spanning fails only when *all three* of the following conditions hold simultaneously: (a) a factor $F^*_{k,t}$ loads *exclusively* on treated countries ($\gamma^*_{c,k} = 0$ for all $c \in \mathcal{C}$); (b) this factor is correlated with the treatment indicator conditional on the spanned factors ($\text{Cov}(F^*_{k,t}, D_{it} \mid \mathbf{F}_t) \neq 0$); and (c) this factor has non-trivial time variation ($\text{Var}(F^*_{k,t}) > 0$). If any one condition fails, the estimator remains consistent. Condition (a) is the most demanding: it requires that a shock affects *only* treated countries, with *no* control country sharing any exposure. Global business cycles, for instance, affect all countries, so CAs capture them regardless. Spanning fails only for shocks that load exclusively on treated countries, a much more specific failure mode than the general heterogeneous-loadings violation that invalidates parallel trends.

Additional assumptions. Our approach imposes four additional assumptions. First, sufficient cross-sectional units in the control sample ($N_c \to \infty$) ensure that CAs are reliable factor proxies. Second, sufficient time periods ($\sqrt{N_{tr}}/T \to 0$) are needed to estimate pair-specific parameters with adequate precision. Third, the observable variables used for CA construction must load on the relevant factors; in practice, gravity fundamentals (income, population) are natural candidates since they enter the structural model directly. Fourth, treatment must not be perfectly correlated with unspanned factors. The CSD diff diagnostic (Section 4.4) provides a direct test of whether the latter two conditions are met in practice.

On the choice of CA variables. The CCE methodology imposes weaker requirements on the variables used to construct cross-sectional averages than structural exogeneity in the sense of a particular economic model. What is required is that the chosen variables (a) load on the underlying common factors, and (b) are not contaminated by treated-country treatments to first order. Country-level income and population, which we use in the empirical application, satisfy both. We highlight this because income is endogenously determined in many structural frameworks (notably structural gravity, where it is solved jointly with the price system as part of the equilibrium), which might appear to disqualify it as a factor proxy. It does not, because the CCE conditions are statistical, not structural: what matters is whether control-country income loads on common shocks and whether it is materially contaminated by treatments elsewhere in the panel. A variable can be endogenously determined within any structural model and still satisfy both conditions.

The contamination question is operational. Control-country income could in principle respond to treatments elsewhere in the panel through trade multipliers: when a treated country's income or trade costs change, its demand shifts, partner exports respond, and partner income moves. The magnitude of this feedback is bounded above by the product of three small factors. First, each treated country's demand shift is spread across up to $N-1$ partners, attenuating the impact on any individual partner by $1/(N-1)$. Second, bilateral exports are a fraction of each partner's total income, attenuating further by the bilateral export share. Third, the cross-sectional average itself averages over N_c control countries, attenuating by $1/N_c$. The combined order is $O(1/((N-1)\cdot N_c))$, which is second-order relative to the $O_p(N_c^{-1/2})$ CA approximation error already accommodated by the asymptotic framework. By contrast, contamination of constructed equilibrium adjustment terms is first-order, because the adjustment terms are defined as functions of all countries' characteristics (each treated country contributes weight $1/N$ rather than $1/((N-1)\cdot N_c)$); this is the formal underpinning of the bad-control concern in Section 3.3. The simulation evidence is direct: in the DGP of Section 4, the income variable responds to both common factors and (in treated countries) to treatment, yet the CCE-DID estimator nevertheless recovers the true ATET with negligible bias.

Total Treatment Effects and Channel Absorption. The CCE-DID estimator recovers the total reduced-form ATET. Under Assumption 1 and as $N_{\mathrm{tr}}, T \to \infty$ with $\sqrt{N_{\mathrm{tr}}}/T \to 0$, the MG estimator is consistent for

$$\theta^{\mathrm{ATET}} = \frac{1}{|M|} \sum_{(i,j,t)\in M} \left(\ln Y_{ij,t}^{\mathrm{tr}} - \ln Y_{ij,t}^{\mathrm{cf}}\right), \tag{12}$$

where M is the set of treated pairs in post-treatment periods, Y^{tr} is the observed outcome, and Y^{cf} is the counterfactual outcome absent treatment.

When variables that capture an intermediate outcome on the causal path are added as regressors, the corresponding channel is absorbed. Including (log) income, for instance, isolates the residual non-income effect, following the terminology of Huber [2014]:

$$\mathbb{E}[Y_{ij,t} \mid \cdot] = \exp\left(\alpha_{ij} + \theta_{ij}^{\mathrm{direct}} D_{it} + \delta_{ij} D_{jt} + \phi_{ij}' \bar{z}_t^{\mathrm{ctrl}} + \beta_{1,ij} \ln z_{it}^{y}\right). \tag{13}$$

This decomposition is valid under the causal ordering $D_{it} \to z_{it}^{y} \to Y_{ij,t}$: income is an intermediate outcome determined by treatment and global shocks before bilateral flows are realised. Including it as a regressor absorbs the income channel, so $\theta_{ij}^{\mathrm{direct}}$ captures the residual treatment effect operating through non-income mechanisms, and $\theta_{ij} - \theta_{ij}^{\mathrm{direct}}$ measures the income-channel contribution. The interpretation requires the causal ordering to be correctly specified and no post-treatment confounders of the income-flow relationship that are not themselves caused by treatment [Cinelli et al., 2024].

Relationship to the existing het-DID literature. The CCE-DID framework subsumes standard het-DID concerns while addressing additional challenges specific to dyadic data. In place of group-time ATTs [Callaway and Sant'Anna, 2021], we estimate pair-specific treatment effects θ_{ij}, disaggregating below the cohort level to accommodate arbitrary heterogeneity across destination partners. The principle of using never-treated controls carries over directly: control-only CAs serve the same role as never-treated comparison groups. We do not use not-yet-treated controls, because including future-treated countries in the CAs introduces contamination analogous to the forbidden comparisons identified by de Chaisemartin and D'Haultfœuille [2020].

Both the CCE-DID and imputation-based estimators [Borusyak et al., 2024] construct counterfactuals from control-group information, but the CCE approach uses a factor model rather than direct outcome imputation, requiring the weaker factor-spanning assumption. Whereas ETWFE [Wooldridge, 2023, Nagengast and Yotov, 2025] handles bilateral treatments varying at the pair level, CCE-DID handles monadic treatments varying at the origin level, where ETWFE cannot identify the effect due to the rank deficiency (Section 3.3). Negative weighting [Goodman-Bacon, 2021] is avoided by construction: each pair's effect is estimated from its own data only, without cross-pair comparisons. Event-study estimation follows the same logic as in Sun and Abraham [2021], applied at the pair level and aggregated via MG.

3.3 Why Standard Modelling Approaches Fail for Monadic Treatments

For bilateral policies that vary at the pair level, such as free trade agreements or bilateral investment treaties, the standard empirical approach uses directional-time fixed effects:

$$\mathbb{E}[Y_{ij,t} \mid \cdot] = \exp\big(\alpha_{ij} + \mu_{it} + \nu_{jt} + \beta \cdot \text{Policy}_{ij,t}\big) \, , \tag{14}$$

where μ_{it} absorbs origin-time unobservables (including the network-wide equilibrium adjustment terms) and ν_{jt} absorbs destination-time unobservables. This is the workhorse specification in the structural-gravity literature[4] and is appropriate for its standard use case: $\text{Policy}_{ij,t}$ varies across pairs within the same origin-time cell, so the FE do not absorb the identifying variation.

[4] For recent surveys see Head and Mayer [2014] and Yotov et al. [2016].

With a monadic treatment, D_{it} varies at the origin-time level and the approach and possible workarounds encounters a number of serious challenges:

(i) Perfect collinearity. D_{it} varies at the origin-time level, exactly as μ_{it} does. Including origin-time fixed effects absorbs D_{it} entirely: the treatment effect is not identified. Formally, with D_i the $T \times 1$ treatment vector for origin i and M_i the $T \times T$ matrix of origin-time dummies, $D_i \in \mathrm{col}(M_i)$, so the treatment coefficient is not estimable. This is a rank deficiency, not a power issue. The standard structural-gravity toolkit cannot accommodate monadic treatments without modification: directional-time FE remain the right tool for bilateral treatments, but they cannot deliver identification when treatment varies at exactly their level of variation.

(ii) Workaround: Bad control and unspanned factor confounding. The collinearity in (i) calls for a workaround. The natural alternative is to replace the directional-time FE with constructed equilibrium adjustment terms used as continuous controls, leaving D_{it} free to enter the regression. This approach fails on two distinct grounds, as confirmed in our simulations:

First, bad control. The constructed adjustment terms are post-treatment objects. They are nonlinear functions of every country's primitive characteristics: in our empirical example, $D_{it} = 1$ raises origin income z^y_{it} (income channel) and lowers bilateral costs $\tau_{ij,t}$ (cost channel), and both feed into the system-wide equilibrium, shifting the adjustment terms for every country. Conditioning on these terms therefore partials out the channels through which treatment operates, the textbook bad-control problem [Cinelli et al., 2024]. On its own, this mechanism attenuates the total reduced-form effect toward zero.

Second, unspanned factor confounding. The constructed proxies do not in general span the underlying common factor space. Pair-level loadings are heterogeneous (a global shock affects pairs of large open economies differently from pairs of small closed economies, with the response depending on both countries' characteristics and on their position in the network of flows), so the residuals retain factor structure that the proxies fail to absorb. This leaves residual confounding correlated with treatment timing.

The two failure modes combine. Mediator attenuation pulls the estimate toward zero; unspanned factor confounding pulls it in whatever direction the omitted factor structure leans. In our simulations (Section 4), factor confounding dominates and the net deviation is $+0.109$, overestimating the true effect by 80%. The specification recovers neither the total effect nor a clean partial effect. We provide a more formal discussion below.

(iii) Heterogeneous loadings and parallel trends. A common time fixed effect cannot rescue the situation, because the equilibrium response is pair-specific. The standard tool for absorbing country-time-specific variation is directional-time fixed effects, which reintroduces the collinearity problem in (i). The two challenges are interlinked: resolving the network problem through fixed effects destroys identification, while ignoring the network heterogeneity invalidates parallel trends. Formally, consider a standard DID comparison without directional-time fixed effects. The identifying assumption in a log-linear model is

$$\mathbb{E}[\ln Y_{ij,t}(0) - \ln Y_{ij,t-1}(0) \mid i \in \text{treated}] = \mathbb{E}[\ln Y_{ij,t}(0) - \ln Y_{ij,t-1}(0) \mid i \in \text{control}], \quad (15)$$

which holds only if factor loadings are homogeneous across treated and control pairs. Under heterogeneous loadings, the standard 2WFE estimator yields

$$\hat{\theta}^{\text{DID}} \xrightarrow{p} \bar{\theta} + \frac{\sum_{ij,t} \widetilde{D}_{it} \, (\lambda_{ij} - \bar{\lambda})' \widetilde{F}_t}{\sum_{ij,t} \widetilde{D}_{it}^2}, \quad (16)$$

where $\widetilde{D}$ and $\widetilde{F}$ are time-demeaned. The second term is non-zero when $\text{Cov}(\lambda_{ij}, D_{it}) \neq 0$.

Equilibrium adjustment terms: the mediator structure formalised. Consider the specification that replaces directional-time fixed effects with explicitly constructed equilibrium adjustment terms.[5] Let $\Pi_{i,t}$ and $\Xi_{j,t}$ denote the origin and destination adjustment terms, solved from the observed post-treatment data, and consider

$$\mathbb{E}[Y_{ij,t} \mid \cdot] = \exp\Big(\alpha_{ij} + \theta_{ij}^{\text{partial}} D_{it} + \phi_{1,ij} \ln \Pi_{i,t} + \phi_{2,ij} \ln \Xi_{j,t}\Big). \quad (17)$$

Proposition 1 (Mediator structure of equilibrium adjustment terms)**.** *Suppose treatment D_{it} affects bilateral flows through an income channel ($D_{it} \to z_{it}^y \to Y_{ij,t}$) and a cost channel ($D_{it} \to \tau_{ij,t} \to Y_{ij,t}$). Let $\Pi_{i,t} = h\big(\{z_{k,t}^y\}_{k=1}^N, \{\tau_{kl,t}\}_{k,l}\big)$ be the equilibrium adjustment term for country i, solved from the full system. Then:*

(i) $\Pi_{i,t}(D_{it} = 1) \neq \Pi_{i,t}(D_{it} = 0)$: the adjustment terms shift in response to treatment.

(ii) Conditioning on $\Pi_{i,t}$ and $\Xi_{j,t}$ partials out the income and cost channels: the coefficient

[5] In structural gravity applications, these adjustment terms take the specific form of multilateral resistance terms [Anderson and Van Wincoop, 2003], and the partial-equilibrium estimand of Proposition 1 below corresponds to the $\sigma - 1$ elasticity of bilateral flows with respect to bilateral costs that the structural-gravity literature typically targets and feeds into counterfactual simulations. The argument that follows applies to any model where bilateral flows are determined in a system-wide equilibrium taking country-level characteristics as inputs.

on D_{it} in (17) targets a partial-equilibrium estimand smaller in magnitude than the total reduced-form effect when channels and total share sign, $|\theta_{ij}^{\text{partial}}| < |\theta_{ij}|$. This is the textbook mediator/bad-control mechanism [Cinelli et al., 2024].

Proof sketch. The adjustment terms are nonlinear functions of all countries' incomes and bilateral costs. Since $D_{it} = 1$ moves both z_{it}^{y} (by θ^{z_1}) and $\tau_{ij,t}$ (by θ^{cost}), and both enter $h(\cdot)$, the equilibrium terms respond to treatment, establishing (i). For (ii), conditioning on $\Pi_{i,t}$ and $\Xi_{j,t}$ partials out the variation in $Y_{ij,t}$ that treatment generates through these channels. The remaining coefficient $\theta_{ij}^{\text{partial}}$ captures only the residual treatment variation orthogonal to the equilibrium response. □

The mediator result and simulation evidence. Proposition 1 characterises the mediator channel in isolation, holding factor confounding fixed. In practice, the two failure modes operate together. Whether the resulting coefficient is biased upward, biased downward, or attenuated cleanly depends on which mechanism dominates and on whether factor confounding has been separately controlled. Two cases are worth distinguishing:

Case A: equilibrium adjustment proxies without CCE controls. This is the specification reported as 'Equilibrium-term controls (no CA)' in our simulations (Table 2). Mediator attenuation pushes the coefficient toward zero; unspanned factor confounding pushes it in the direction of the omitted factor structure. In our calibration the latter dominates and the net deviation from the total effect is $+0.109$, recovering neither the total effect nor a clean partial effect. The specification is biased, not informative.

Case B: equilibrium adjustment proxies jointly with CCE controls. This is the empirical "Add eq. terms" specification reported in Section 5.3. With CAs absorbing factor confounding, the residual movement when the adjustment proxies are added comes from two sources. The first is the mediator attenuation predicted by Proposition 1: the MRT-mediated channels of income and cost are partialled out. The second is a structural side-effect of how MRT proxies are constructed: the proxies depend on country-level incomes through GDP shares.[6] Including the proxies therefore also absorbs a portion of the direct size effect of Y_i. The coefficient drops from 0.411 (total effect, benchmark) to 0.146. This residual effect is informative about the combined contribution of MRT-mediated channels and the structurally absorbed portion of direct size, but it should not be interpreted as a clean $\sigma - 1$ trade-cost elasticity in the sense

[6] In the Anderson and Van Wincoop [2003] system, P_j is a function of Y_i via the share weights θ_i; the Baier and Bergstrand [2009] linearisation has the same structural dependence.

of the structural-gravity literature. Recovering such an elasticity in a clean form would require additional structure (for instance, jointly conditioning on Y_i and on the residual MRT variation orthogonal to Y_i), which is beyond the scope of this paper.

The contrast between Cases A and B clarifies the relationship between our methodology and standard structural practice. For *bilateral* treatments, standard structural gravity uses directional-time FE, which absorb the adjustment terms perfectly and cleanly identify the partial trade-cost effect. For *monadic* treatments, directional-time FE cannot be used, and the standard workaround (equilibrium adjustment proxies on their own) fails because factor confounding remains (Case A). Adding CCE controls *alongside* the proxies (Case B) controls for factor confounding and partials out the MRT-mediated channels, but the residual coefficient also reflects partial absorption of the direct size effect through the proxies' dependence on GDP shares. Case B therefore yields an informative residual rather than a clean $\sigma - 1$ elasticity; a clean structural decomposition would require additional restrictions that we do not impose. Using CCE controls alone, without the equilibrium adjustment proxies, recovers the total reduced-form effect that this paper targets.

3.4 Treatment Reversal and Carryover Effects

In many relevant applied contexts, treatments are reversed (e.g., democracy collapses) or are short-lived to begin with (e.g., periods of 'democratic backsliding', a decline in institutional quality short of democratic collapse). The existing het-DID literature largely assumes *absorbing* treatments and does not accommodate treatment reversal.[7] The natural question in this context is: what does the static coefficient $\hat{\theta}_{ij}$ identify when D_{it} switches on and off again?

The estimand. The CCE-DID estimator regresses bilateral flows on the binary treatment indicator D_{it}, pair fixed effects, and cross-sectional averages. The Mean Group coefficient aggregates pair-level estimates of

$$\hat{\theta}_{ij} \approx \overline{Y_{ij,t}}^{D=1} - \overline{Y_{ij,t}}^{D=0}, \tag{18}$$

where the averages are over periods when country i is treated ($D_{it} = 1$) versus untreated ($D_{it} = 0$). For countries with multiple transitions, both sums draw from interrupted/non-continuous periods of treatment and non-treatment. The effect of this setup on the estimated

[7]Notable exceptions include Liu et al. [2024] and de Chaisemartin and D'Haultfœuille [2024].

treatment effect $\hat{\theta}_{ij}$ will turn out not to depend on the 'cycling' between treatment and reversal *per se*, but on the existence of 'carryover' effects after the end of the treatment period.

Clean reversal (assuming no carryover). Any total effects of treatment on bilateral flows or on its intermediate channels (income, cost) are contemporaneous and fully reversible: when D_{it} reverts to zero, these outcomes return to their counterfactual path/pre-treatment level. The general equilibrium is re-solved at each period.

Under this specification, D_{it} correctly captures the treatment status at each t. The contrast in (18) compares periods with treatment to periods without, and the estimator recovers the true average treatment effect θ^{true}. Multiple transitions do not threaten identification — they simply provide additional variation for estimation, with each democratic episode contributing to the $D = 1$ mean and each autocratic episode to the $D = 0$ mean.

Reversal with carryover. The key threat is *carryover*: if effects persist after reversal, then $D = 0$ periods retain residual treatment effects from the previous treatment episode(s).[8] Specifically, if a fraction $\rho \in [0, 1]$ of the treatment effect persists per year after $D \to 0$, the effective treatment intensity is

$$E_{it} = \begin{cases} 1 & \text{if } D_{it} = 1 \\ \rho \cdot E_{i,t-1} & \text{if } D_{it} = 0. \end{cases} \tag{19}$$

With $\rho > 0$, the contrast between $D = 1$ and $D = 0$ periods shrinks because $D = 0$ periods still carry residual treatment effects. The average effective treatment during a post-reversal episode of length $\bar{L}$ is and treatment persistence/carryover ρ is

$$\frac{1}{\bar{L}} \sum_{k=1}^{\bar{L}} \rho^k = \frac{\rho(1 - \rho^{\bar{L}})}{\bar{L}(1 - \rho)}, \tag{20}$$

leading to attenuation bias. Table 1 illustrates the effective treatment for post-reversal episodes up to a dozen years and persistence ranging from 0.1 to 0.9.

Empirical relevance. Whether carryover from reversal is a concern depends on the nature of the treatment effects and the dynamics of causal interaction. Informally, the subsample of countries with a single treatment and no reversal provides estimates that are uncontaminated

[8]For instance, positive effects from treatment may spill over to the period after reversal, negative effects may result in a 'hangover' from treatment or a 'bounce back' overshooting the pre-treatment equilibrium level.

by carryover and can serve as a benchmark: if carryover is strong in the application at hand, the full-sample estimates would be substantially attenuated toward zero relative to this benchmark.

Table 1: Effective treatment during post-reversal episodes under carryover

	ρ								
$\bar{L}$	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
1	0.100	0.200	0.300	0.400	0.500	0.600	0.700	0.800	0.900
2	0.055	0.120	0.195	0.280	0.375	0.480	0.595	0.720	0.855
3	0.037	0.083	0.139	0.208	0.292	0.392	0.511	0.651	0.813
4	0.028	0.062	0.106	0.162	0.234	0.326	0.443	0.590	0.774
5	0.022	0.050	0.086	0.132	0.194	0.277	0.388	0.538	0.737
6	0.019	0.042	0.071	0.111	0.164	0.238	0.343	0.492	0.703
7	0.016	0.036	0.061	0.095	0.142	0.208	0.306	0.452	0.671
8	0.014	0.031	0.054	0.083	0.125	0.184	0.275	0.416	0.641
9	0.012	0.028	0.048	0.074	0.111	0.165	0.249	0.385	0.613
10	0.011	0.025	0.043	0.067	0.100	0.149	0.227	0.357	0.586
11	0.010	0.023	0.039	0.061	0.091	0.136	0.208	0.332	0.561
12	0.009	0.021	0.036	0.056	0.083	0.125	0.192	0.310	0.538

Notes: We compute the effective treatment for a post-reversal episode of length $\bar{L}$ and persistence ρ, using the formula in equation (20).

Formal testing. We take our lead from Liu et al. [2024], who devise a diagnostic test for carryover effects in two-way panels with staggered multiple treatments and reversals, analysing the periods immediately *after* treatment reversal. In the dyadic flow data context, let $R_{it}^{(k)} = \mathbf{1}\{t = T_i^{\text{off}} + k\}$ denote an indicator for being k periods after treatment reversal, which is added to the pair-level treatment regression for exporters which experienced treatment reversal(s):

$$\mathbb{E}[Y_{ij,t} \mid D_{it}, D_{jt}, \bar{\mathbf{z}}_t^{\text{ctrl}}] = \exp\left(\alpha_{ij} + \theta_{ij} \cdot D_{it} + \delta_{ij} \cdot D_{jt} + \boldsymbol{\phi}_{ij}' \bar{\mathbf{z}}_t^{\text{ctrl}} + \sum_{k=1}^{K} \gamma_{ij,k} R_{it}^{(k)} + \kappa_{ij} PR_{it}\right), \quad (21)$$

where $PR_{it} = \mathbf{1}\{t \geq T_i^{\text{off}} + K + 1\}$ represents all post-reversal time periods *after* the K periods tested for carryover. Since pre-treatment periods are the omitted category, including PR_{it} ensures treatment and carryover effects are measured relative to pre-treatment, not a mix of pre-treatment and a share of post-reversal observations. The null hypothesis of no carryover effects is then $H_0 : \gamma_{ij,1} = \gamma_{ij,2} = \cdots = \gamma_{ij,K} = 0$, which can be tested using a joint Wald test of the Mean Group estimates $\hat{\gamma}_1, \hat{\gamma}_2, \ldots, \hat{\gamma}_K$.

Practical implications. In empirical applications where carryover is apparent (Wald test rejects), any *full sample* ATETs will be distorted (reversal is not 'clean'), but we can readily quantify the trade effects of different regime stints. If a subset of countries experiences a back and forth of treatment and reversal, as is the case in our empirical application of democratic regime change, which violate the no-carryover assumption, then our heterogeneous treatment estimation approach enables us to compute treatment effects by country 'type': first, for countries with a single (absorbing) treatment, we do not need to concern ourselves with carryover since their ATET or treatment event time estimates are unaffected; second, for countries with treatment reversal, adopting the regression in equation (21) maintains the identifying assumption that only pre-treatment observations are the omitted benchmark category, whereas all post-reversal observations are separately captured/identified via the carryover dummies, $R^{(k)}$, and the PR dummy for all post-reversal periods after the K carryover periods; third, for countries with more complex regime dynamics we can, degrees of freedom permitting, further add respective treatment dummies for a second, third, etc. stint in democracy, as well as carryover dummies for a second, third, etc. autocratic reversals. We illustrate this in our empirical application in Section 5 by estimating the respective average treatment effect (per year of treatment) for exporters with different democratic regime change histories during our sample period.

4 Finite-Sample Performance and Diagnostics

We conduct a large-scale simulation study to examine the finite-sample properties of the CCE-DID estimator. Full details of the data-generating process and all configurations are provided in Appendix B; here we summarise the design and key results. Our DGPs are inspired by our application, the export effects of democratic regime change.

4.1 Data-Generating Process

The DGP generates bilateral flows from a structural model with N countries and T periods. Two AR(1) common factors $\mathbf{F_t}$ (with AR coefficient $\rho = 0.7$) drive country-level fundamentals:

$$\ln z_{1,it} = \alpha_i^{z1} + \boldsymbol{\gamma}_i^{z1\prime}\mathbf{F}_t + \theta_{z_1} \cdot D_{it} + u_{it} \tag{22}$$

$$\ln z_{2,it} = \alpha_i^{z2} + \boldsymbol{\gamma}_i^{z2\prime}\mathbf{F}_t + \xi_{it} \tag{23}$$

where z_1 (income) responds to treatment ($\theta_{z_1} = 0.13$) and z_2 (population) does not. Factor loadings γ_i are drawn heterogeneously across countries. The fundamental loadings are correlated ($\gamma_i^{z_2} = 0.3 \cdot \gamma_i^{z_1} + \nu_i$), providing additional factor-spanning power through the cross-loading structure.

Bilateral costs also follow a factor structure:

$$\ln \tau_{ij,t} = \ln \bar{\tau}_{ij} + \gamma_i^{\tau\prime} \mathbf{F}_t + \gamma_j^{\tau\prime} \mathbf{F}_t + \frac{\theta_{\text{cost}}}{1-\sigma} \cdot D_{it} \tag{24}$$

with $\theta_{\text{cost}} = 0.10$ and $\sigma = 5$ (the trade elasticity).

Treatment selection (i.e., which countries democratise) is random, but treatment timing is correlated with the global factor $F_{1,t}$: countries tend to democratise during global expansions [Li and Reuveny, 2003, Milner and Mukherjee, 2009, Gunitsky, 2014], with $\text{Corr}(D_{it}, F_t) \approx 0.2$–$0.4$ (equation (34) in Appendix B). This induces a selection-on-timing pattern: countries democratising during global expansions experience contemporaneous factor-driven GDP and MRT shifts, implying that any estimator ignoring heterogeneous factor exposure will be biased.

The system equilibrium is solved numerically via fixed-point iteration [Fally, 2015], ensuring internal consistency. The true total reduced-form ATET is computed by comparing actual flows to a counterfactual world in which no country receives treatment (re-solving the equilibrium at each t). The partial-equilibrium prediction of the treatment effect, which holds equilibrium adjustment terms constant, is

$$\theta^{\text{PE}} = \theta_{z_1} + \theta_{\text{cost}} = 0.13 + 0.10 = 0.23 \tag{25}$$

However, general-equilibrium forces attenuate this, yielding $\theta^{\text{true}} \approx 0.137$ at $N = 60$, a GE attenuation of approximately 40%.

The *baseline* scenario has all countries loading on both factors so that control CAs span the factor space; this is the benchmark for assessing bias and coverage. The *treated factor* scenario introduces an additional factor that loads exclusively on treated countries, is trending, and is correlated with treatment timing, thereby creating a deliberate spanning failure. Its strength is parameterised by $s \in [0, 1]$: at $s = 0$ the factor is absent (the baseline), $s = 0.5$ represents moderate confounding, and $s = 1$ produces extreme confounding where the unspanned factor dominates treated-country variation.

4.2 Key Results

Main Results. Table 2 summarises the main simulation findings, adopting the rreg Mean Group aggregation. Control-only CAs deliver negligible bias ($+0.004$) with RMSE of 0.011, confirming that the estimator recovers the true ATET. Alternative strategies produce economically large bias: no CAs overestimates by 111%, dual CAs underestimate the treatment effect by 42%, and a specification replacing CAs with equilibrium-term proxies overestimates by 80%, driven by residual factor confounding that the equilibrium terms fail to absorb.

Table 2: Simulation Results: Estimator Comparison

Specification	Bias	RMSE	Coverage
CA strategy (baseline scenario)			
Control-only CAs	$+0.004$	0.011	94%
No CAs	$+0.152$	0.179	–
Dual CAs (all countries)	-0.057	0.059	–
Equilibrium-term controls (no CAs)	$+0.109$	0.112	–
Robustness (control-only CAs)			
Heterogeneous effects ($\sigma_{\text{het}} = 0.1$)	$+0.003$	0.028	–
Heterogeneous effects ($\sigma_{\text{het}} = 0.2$)	-0.009	0.050	–
Factor misspecification (3 factors, 2 CAs)	$+0.004$	0.024	–
Factor misspecification (4 factors, 2 CAs)	-0.009	0.032	–
Zero trade flows (30% of pairs)	$+0.000$	0.008	–
Spanning failure (treated factor)			
Intermediate confounding ($s = 0.5$)	$+0.060$	0.080	–
Extreme confounding ($s = 1.0$)	$+0.202$	0.244	–
CSD diff diagnostic	See Table 3		

Notes: 15 replications per configuration. Baseline calibration ($N = 60$, $T = 65$, 50% treated). True ATET ≈ 0.137. Two-way clustered SEs [Cameron et al., 2011]. Dual CAs compute cross-sectional averages over *all* countries (treated and control), introducing treatment contamination as described in equation (6). The equilibrium-term specification replaces CAs with MRT proxies and is exposed to residual factor confounding.

Robustness. The estimator is also robust to misspecification: MG aggregation handles arbitrary treatment effect heterogeneity, and two CA variables per country dimension suffice even when the DGP has 3 or 4 factors. With 30% zero trade flows (the extensive margin), the bias remains negligible, confirming that the PPML estimator handles zeros without distortion.

Spanning failure. When an unspanned factor loads exclusively on treated countries ($s > 0$), the control CAs can no longer absorb it and bias increases with s. At moderate confounding ($s = 0.5$), the bias reaches $+0.060$, economically meaningful at 44% of θ^{true}, while at extreme confounding ($s = 1.0$) it rises to $+0.202$. This motivates the CSD diff diagnostic (see below).

4.3 Further Simulation Results

In Appendix B we present a range of additional simulation results which we briefly summarize here. Estimator performance is stable across sample sizes from $N = 30$ exporters to $N = 160$ (Appendix Table B-1). We can also introduce more intricate dependencies, related to treatment (e.g., if treated countries are richer than control one) and weaker factor loadings in the control sample. Appendix Table B-2 demonstrates that these scenarios do not lead to meaningful bias relative to our baseline specification. Finally, existing work on the structural gravity model of bilateral trade flows has emphasised the importance of incorporating domestic trade flows [e.g., Yotov, 2022]. In Appendix Table B-3 we show that excluding theory-consistent domestic trade flows does not alter the performance of the CCE-DID estimator.

4.4 Factor Spanning Diagnostic: CSD Diff

We develop a diagnostic to assesses whether Assumption 1 holds in a given application, building on differences in average residual correlations between treated and control samples (ΔCSD). The core idea exploits a key implication of adequate spanning: if CAs successfully absorb the common factor structure for both treated and control pairs, their residuals should exhibit similar levels of cross-sectional dependence (CSD), $\Delta\text{CSD} \approx 0$. If an unspanned factor loads on treated countries and is correlated with treatment, it generates systematic residual structure among treated pairs that is absent among control pairs, producing $\Delta\text{CSD} > 0$.

Test statistic. Let $\hat{e}_{ij,t}$ denote the Pearson residual from the pair-level PPML regression (8). Define the mean absolute pairwise correlation of residuals within group g:

$$\overline{|\rho|}_g = \frac{2}{n_g(n_g - 1)} \sum_{\substack{(i,j)<(k,l) \\ (i,j),(k,l)\in g}} \left|\text{corr}(\hat{e}_{ij,\cdot}, \hat{e}_{kl,\cdot})\right|, \qquad g \in \{\text{treated}, \text{control}\} \tag{26}$$

The CSD diff statistic is then:

$$\Delta\text{CSD} = \overline{|\rho|}_{\text{treated}} - \overline{|\rho|}_{\text{control}} \tag{27}$$

Null hypothesis. Under H_0 Assumption 1 on adequate spanning holds, so $\mathbb{E}[\Delta\text{CSD}] = 0$. Under the alternative (unspanned factors load on treated pairs), treated-pair residuals retain systematic factor structure, yielding $\Delta\text{CSD} > 0$.

Permutation inference. Exact p-values are obtained by permuting origin treated/control labels $B = 999$ times and recomputing ΔCSD under the null, following the $(b + 1)/(B + 1)$ formula of Phipson and Smyth [2010]. The permutation is performed at the origin level (not the pair level), preserving the within-origin correlation structure and ensuring valid inference despite the dyadic dependence.

Simulation performance. Table 3 reports the CSD diff diagnostic across three levels of confounding strength. At $s = 0$ (good spanning, no confounding): $\Delta\text{CSD} \approx -0.000$ with $p = 0.520$, correctly indicating no spanning failure. At $s = 0.5$ (moderate confounding): $\Delta\text{CSD} \approx 0.080$ with $p = 0.026$, where the bias in the benchmark model with control-country CA already reaches $+0.06$ (43% of θ^{true}). At $s = 1.0$ (extreme confounding) the bias is catastrophic ($+0.202$), but ΔCSD is only 0.041 with $p = 0.310$. The unspanned factor is so strong that it overwhelms both groups' residuals: treated pairs have large residual CSD from the unspanned factor, but control pairs also have elevated residual CSD because the CAs (which no longer span the full factor space) leave more residual structure for everyone. The differential shrinks even though absolute CSD rises for both groups. The diagnostic therefore has strong power in the moderate range where early detection matters most.

Table 3: CSD Diff Diagnostic Across Spanning Quality

Confounding strength s	ΔCSD	Mean p-value	Estimator bias	Bias/ θ^{true}
$s = 0$ (good spanning)	-0.000	0.520	$+0.004$	3%
$s = 0.5$ (moderate confounding)	$+0.080$	0.026	$+0.060$	44%
$s = 1.0$ (extreme confounding)	$+0.040$	0.310	$+0.202$	155%

Notes: ΔCSD is the difference in mean absolute pairwise residual correlation between treated and control pairs. p-values from 999 exporter-level permutations. We report the bias in the control country-only CA CCE-DID for each scenario and Bias/θ^{true} from the same simulation draw.

4.5 Treatment Reversal and Carryover

We extend our simulations with two new DGP variants that generate bilateral flows responding to 'cycling' treatment patterns to analyse the effect of ignoring treatment reversal.

Multiple Transition DGP. A share r of treated countries experiences reversals by cycling through on/off/on episodes, each lasting L years. The treatment indicator is binary and matches the true DGP: when $D_{it} = 0$, trade costs and GDP revert to their pre-treated values, and the general equilibrium is re-solved. The other $1-r$ treated countries receive permanent treatment.

Carryover DGP. The same cycling binary indicator D_{it} is used, but the *effective treatment* entering the DGP is $E_{it} \in [0,1]$ with persistence parameter ρ. When $D_{it} = 1$, $E_{it} = 1$; when $D_{it} = 0$, $E_{it} = \rho \cdot E_{i,t-1}$ (exponential decay): treatment carries over. Trade costs and GDP respond to E_{it}, but the estimator sees only D_{it}. This creates a wedge between the regression indicator and the actual treatment intensity, producing attenuation bias proportional to ρ.

Calibration. Both DGPs use the baseline simulation parameters ($N = 60$, $T = 65$, 50% treated, $\theta_{\text{cost}} = 0.10$, $\theta_{\text{GDP}} = 0.13$, $\sigma = 5$). The cycling share is $c \in \{.43, .50, .73\}$, with $n = 2$ additional transitions after initial onset and episode length $L = 12$ years. The carryover parameter varies across $\rho \in \{0.3, 0.7, 1.0\}$.

Table 4: Treatment Reversal and Carryover Simulations

Test	Control CAs (Bias)	No CAs (Bias)	Dual CAs (Bias)	MRT FE (Bias)
Multiple transitions (true cycling, no carryover)				
50% cycle, 2 trans, 15yr	−0.003	+0.118	−0.034	+0.103
73% cycle, 2 trans, 12yr	−0.005	+0.111	−0.035	+0.101
73% cycle, 4 trans, 8yr	−0.003	+0.103	−0.026	+0.104
Hysteresis (persistent carryover after reversal)				
30% persistence	+0.018	+0.134	−0.014	+0.123
70% persistence	+0.007	+0.125	−0.029	+0.109
100% persistence	−0.047	+0.075	−0.102	+0.035

Notes: 15 replications per configuration. $N = 60$, $T = 65$, 50% treated, baseline specification. True ATET: $\theta^{\text{true}} \approx 0.134$ for clean cycling (upper panel), 0.110 for $\rho = 0.3$, 0.114 for $\rho = 0.7$, 0.137 for $\rho = 1.0$. The true ATET is computed as the average noiseless flow difference between actual (with treatment) and counterfactual (no treatment), over $E_{it} > 0$ periods. Cycling share $c = 0.43$ with 2 additional transitions and 12-year episodes, except in the first and third row of the upper panel (respectively, $c = 0.50$, 15yr and $c = 0.73$, 4 transitions, 8yr).

Results. Table 4 reports bias for each estimator across six configurations. Experiments in the upper panel demonstrate that in our preferred specification with control-sample CA the estimator remains unbiased when treatment cycles on and off: bias is -0.003 to -0.005, negligible relative to $\theta^{\text{true}} \approx 0.134$. The structural gravity model is static in the sense that trade costs respond contemporaneously to D_{it}: when a country reverts to autocracy, trade barriers return to their pre-treatment level, and the general equilibrium adjusts accordingly. The binary D_{it} correctly captures the treatment status at each period, and the MG coefficient recovers the true ATET regardless of whether countries have one or more transitions.[9] This result holds because the CCE-DID estimator exploits within-pair variation over time. Multiple transitions provide additional on/off contrasts within each pair's time series, which — under the correct specification — improve precision rather than introducing bias.

Experiments in the lower panel show that the threat to identification is not cycling *per se* but *persistence*: if the trade effect of democracy survives autocratic reversal, the binary indicator D_{it} understates the true treatment intensity during autocratic episodes. With moderate to large persistence ($\rho = 0.3$–0.7), the bias remains small ($+0.007$ to $+0.018$) because the true ATET itself is reduced by carryover (the counterfactual comparison accounts for the persistent effective treatment). At full irreversibility ($\rho = 1.0$), the bias reaches -0.047, reflecting the complete decoupling between the binary indicator and the actual treatment intensity: once a country has been democratic, the trade effect persists permanently, and the on/off contrast in D_{it} becomes uninformative about the true treatment status.

5 Application: Democracy and Bilateral Exports

5.1 Background

Democratic regime change is a prominent example of a monadic treatment with expected effects on bilateral flows. Within the political theory of trade, a 'direct democracy' model suggests that democratic regimes act, under majority rule, in agreement with the trade preferences of the median voter [Mayer, 1984, Dutt and Mitra, 2002]. In a standard Heckscher-Ohlin framework, assuming that countries undergoing democratisation are labour-abundant, capital-poor workers gain from trade liberalisation; extension of the franchise implies that the median voter is a worker, and therefore greater trade openness is adopted by the newly elected

[9] Alternative specifications without CA, using dual CA, or MRTs, all display bias, although this bias is *attenuated* for the former two compared with their performance in the benchmark DGP in Table 2.

government. A 'political support' or 'protection for sale' approach [Grossman and Helpman, 1994, Rodrik, 1995] would also allow existing interest groups to lobby for their preferred trade policies, but their political influence is likely to be substantially eroded in the newly democratised regime [Milner and Mukherjee, 2009].

In addition, democratisation can have indirect economic effects by promoting legal and economic reforms which support all businesses [Rode and Gwartney, 2012, Giuliano et al., 2013]. The indirect income channel, whereby democracy causes economic growth [Acemoglu et al., 2019], generates higher bilateral flows through the gravity mechanism. Democracy's growth effect emerge in stages [Boese-Schlosser and Eberhardt, 2024], suggesting dynamic treatment effects that our event-study approach is well suited to capture.

The empirical literature on the effects of democratisation on trade is limited and typically deviates from the prescribed model specification of structural gravity. Overall, it finds a positive impact of democracy on the imports and exports of developing countries [Milner and Kubota, 2005, Eichengreen and Leblang, 2008, Kono, 2008, Tavares, 2008, Aidt and Gassebner, 2010, Yu, 2010]. We provide the first credible causal estimates by addressing the fundamental identification challenges of monadic treatments in dyadic flow data.

5.2 Data

Bilateral exports. We use TRADHIST [Fouquin and Hugot, 2016] covering bilateral exports for 1950–2014 in levels (including zeros; we further code 'plausible zeros' as zero).

Democracy. Motivated by Mukand and Rodrik [2020] we use liberal democracy as defined by the V-Dem index [Coppedge et al., 2021], dichotomised at the full-sample mean. Treatment $D_{it} = 1$ if country i's V-Dem score exceeds the threshold (0.34) in year t.[10] In this baseline specification, 86 countries experienced democratic regime change (at least once) during the sample period. Robustness checks (i) add or subtract fractions of the full sample standard deviation to the V-Dem index mean to define alternative democracy cut-offs, or (ii) adopt the Polity dataset [Marshall et al., 2017, v.5] to define democracy as a polity2 score in excess of 5.

Cross-sectional averages. Annual CAs are constructed from a strict never-treated control sample of 50 countries over 1950–2014. The CAs include $\ln$ GDP and $\ln$ POP for exporter

[10] This follows the practice in recent applied work [e.g., Cho et al., 2026, Sima, 2026] defining regime change as significant and persistent improvements in the underlying index score.

and importer dimensions, taken from TRADHIST. These never-democratic countries are heterogeneous, spanning oil exporters, East Asian economies, and former communist states, which supports the spanning assumption: their diverse factor loading structures make it unlikely that a common shock would load exclusively on democratising countries while leaving each and every one of the 50 control countries unaffected.

Dyadic controls. CEPII data on free trade agreements (FTA) and common currency.

Income controls. For specifications decomposing total into direct effects: $\ln$(GDP per capita) for origin and destination from TRADHIST.

Sample. We discard country pairs with fewer than 14 observations (0.1%) — this is by the necessity of the demands of our heterogeneous panel estimator. The estimation sample contains almost 9,000 pairs in the benchmark specification, with 87 (rreg 86) treated exporters.[11]

5.3 Average Treatment Effects

Table 5 reports ATET estimates across specifications organised in three panels. Panel A presents the main results; Panel B tests robustness to alternative CA constructions, control sets, and definitions of democracy; Panel C provides 'proof-of-concept' specifications that speak to the estimator's theoretical predictions. For the rreg aggregation we report the standard errors based on two-way clustering by exporter and importer [Cameron et al., 2011]. In the MG aggregation the same approach produces degenerate standard errors since taking the median in the first step eliminates most of the within-cluster variation. We therefore do not report these.

Main results. The benchmark specification yields an ATET of 0.411 (rreg, highly significant) or 0.440 (MG). In economic terms, countries that democratise experience on average 50% higher bilateral exports ($\exp(0.411) - 1 \approx 51\%$). This is the *total* treatment effect inclusive of income, cost effects and equilibrium effects, after accounting for dyadic controls (RTAs,

[11] Our sample contains (a) 53 (rreg 49) exporters which experience regime change once and remain democratic for the remainder of our sample period; (b) 34 (rreg 29) exporters which revert to autocracy after earlier democratic regime change (or experience more complex regime change dynamics); (c) 3 countries (ARM, BLR, CUB) which start out as democratic and experience democratic collapse during our sample period (discarded); (d) 28 countries which are always democratic (discarded); and (e) 50 countries which are always autocratic (control). Our aggregation employing robust regression marginally changes these figures, as indicated in parentheses above: pairwise estimates for a small number of countries are extreme outliers and assigned zero weight, likely due to very few pre-treatment or in-treatment observations.

Table 5: Average Treatment Effect on the Treated (ATET)

	Benchmark		Add GDPpc			
Specification	rreg	MG	rreg	MG	Pairs	$G^{\dagger}$
Panel A — Main						
Benchmark	0.411***	0.440	0.105***	0.103	8,986	86
	(0.064)		(0.026)			
Single regime change	0.624***	0.762	0.178***	0.187	4,905	49
	(0.104)		(0.041)			
Panel B — Robustness						
GDP-only CAs	0.283***	0.260			8,992	86
	(0.060)					
No dyadic controls	0.449***	0.488			9,024	86
	(0.068)					
LibDem mean – 1/4 SD	0.563***	0.574			9,329	93
	(0.082)					
LibDem mean + 1/4 SD	0.312***	0.443			7,879	74
	(0.066)					
LibDem mean + 1/2 SD	0.300***	0.451			6,622	62
	(0.074)					
PolityV > 5	0.357***	0.248			8,055	81
	(0.068)					
Panel C — Proof of concept						
Impose Unit Mass	−0.042	−0.000			5,488	84
	(0.027)					
DepVar CA (all)	0.232***	0.253			8,967	86
	(0.054)					
DepVar CA (control)	0.186***	0.247			8,975	86
	(0.054)					
Add eq. terms	0.146***	0.177			8,940	86
	(0.034)					
Diagnostics ‡ (Panel A only)						
CSD diff	ΔCSD $= -0.004$	$p = 0.623$			8,986	86
Carryover	$p<0.001$		p=0.035		2,203	29

Notes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Standard errors (in parentheses) use two-way clustering by exporter and importer [Cameron et al., 2011] with $t(\min(G_{ex}, G_{im}){-}1)$ critical values. Because taking the median in the first step eliminates most within-cluster variation the CGM standard errors for the MG aggregation are degenerate and hence not reported. *Benchmark*: pair-level PPML with pair FE, FTA, common currency, and control-only CAs (GDP + POP from 50 never-democratic countries). *Add GDPpc*: adds $\ln$(GDP per capita) for origin and destination. *Single regime change*: restricts to exporters with exactly one democratisation and no reversal. *GDP-only CAs*: drops population CAs. *No dyadic controls*: drops FTA and common currency. *LibDem ±1/4 SD, LibDem +1/2 SD*: adopts alternative definitions of democratic regime change. *PolityV > 5*: adopts the Marshall et al. [v. 5, 2017] data to define democracy. *Add Unit Mass*: drops $\ln$GDPpc for exporter and destination but divides bilateral flows by origin $\times$ destination GDP (unit income elasticity). *DepVar CA (all/control)*: adds cross-sectional average of $\ln$(flows) from all/control pairs. *Add eq. terms*: adds equilibrium adjustment proxies as regressors. †G denotes the number of exporter clusters with non-zero robust-regression weight. Most specifications retain 83–86 exporters. The single-regime-change specification restricts to exporters with exactly one transition ($G = 49$). The Unit Mass specification has fewer convergent pairs because dividing flows by GDP amplifies measurement noise in small economies. ‡The CSD diff tests the null of no factor spanning failure. The carryover test the null of no significant hysteresis in the treatment effect (in pairs where exporters experienced reversal).

common currency unions) — we report dynamic treatment effects in Section 5.4. When GDP per capita is included to absorb the income channel, the effect drops sharply to 0.105 (rreg) and 0.103 (MG) — we discuss the quantification of the (indirect) income effect in Section 5.5.

The single-regime-change specification restricts the sample to exporters with exactly one democratisation event and no subsequent reversal (49 exporters, 4,905 pairs), eliminating any confounding from treatment reversals. The benchmark ATET rises to 0.624 (rreg, 86%) and 0.762 (MG, 114%), suggesting that irreversible democratisations produce substantially larger trade gains than the full sample average, which includes countries with multiple transitions and reversals potentially attenuating the treatment effect. With GDP per capita included, the direct effect is then 0.178 (rreg) and 0.187 (MG). We discuss capturing a 'cleaner' average treatment effect by regime reversal dynamics in Section 5.5 below.

Robustness. These specifications test sensitivity to CA construction, control variables, and sample period. GDP-only CAs (dropping population) yield 0.283/0.260 (rreg/MG), below the baseline results, suggesting that population CAs provide additional spanning power. Omitting dyadic controls (FTA, common currency) raises the ATET ever so slightly to 0.449/0.488, indicating that these bilateral confounders play a limited role. Defining democracy by adopting the mean cut-off for the V-Dem Liberal Democracy index is somewhat arbitrary, we therefore construct more liberal or restrictive definitions using the mean minus 1/4 (0.563), plus 1/4 (0.312) and plus 1/2 (0.300) of a standard deviation, respectively. Finally, we use a different data source [Marshall et al., 2017, v. 5] and, following these researchers' recommendation, adopt a threshold value of 5 to define democracy (0.357). Across all robustness checks, the ATET remains positive, statistically significant, and economically substantial.

Proof of concept specifications. These specifications validate the estimator's theoretical predictions (see Appendix A), which we had emphasised in the simulation study:

Add Unit Mass: Dividing flows by the product of origin and destination GDP imposes that the income elasticity of trade is exactly unity. In a structural gravity model, the theoretical elasticity of bilateral trade with respect to GDP is 1 in the cross-section, but the effective within-pair elasticity in a panel with pair fixed effects can differ because general equilibrium adjustments (multilateral resistance shifts) absorb part of the income variation. If the imposed elasticity exceeds the effective within-pair elasticity, dividing by GDP absorbs part of the treatment effect operating through non-income channels. The (near-)zero estimates (-0.042

rreg, -0.000 MG) are consistent with this over-correction.

DepVar CAs: CAs constructed from bilateral flows rather than fundamental variables inherit treatment contamination through the equilibrium adjustment channel, as demonstrated in our simulation analysis (Section 3.1). Adding all-pair DepVar CAs yields 0.232/0.253, while control-only DepVar CAs produce 0.186/0.247. Both fall substantially below the benchmark of 0.411/0.440, consistent with our simulation prediction. Even CAs computed exclusively from never-treated control pairs are contaminated because bilateral flows reflect the general equilibrium response to treatment elsewhere in the network: when a country democratises and its exports expand, its trading partners' import patterns shift, which propagates into the cross-sectional averages of control-pair flows.

Add equilibrium terms: Including equilibrium adjustment proxies as regressors alongside CAs produces a residual partial estimate of 0.146 (rreg) and 0.177 (MG), 60–65% below the benchmark $0.411/0.440$. Consistent with Case B in Section 3.3, the drop reflects two ingredients operating jointly: the mediator attenuation formalised in Proposition 1 (the MRT-mediated channels of income and cost are partialled out) and the structural absorption of a portion of the direct size effect of Y_i via the MRT proxy's dependence on country-level incomes. The residual is informative about the combined magnitude of these two effects, but it is not a clean recovery of the structural $\sigma - 1$ trade-cost elasticity. A structural decomposition into MRT-mediated and cost components would require additional restrictions that we do not impose.

Taken together, these results support the theoretical predictions from the simulation study and reinforce the importance of constructing CAs from fundamental variables rather than from equilibrium objects or bilateral flows.

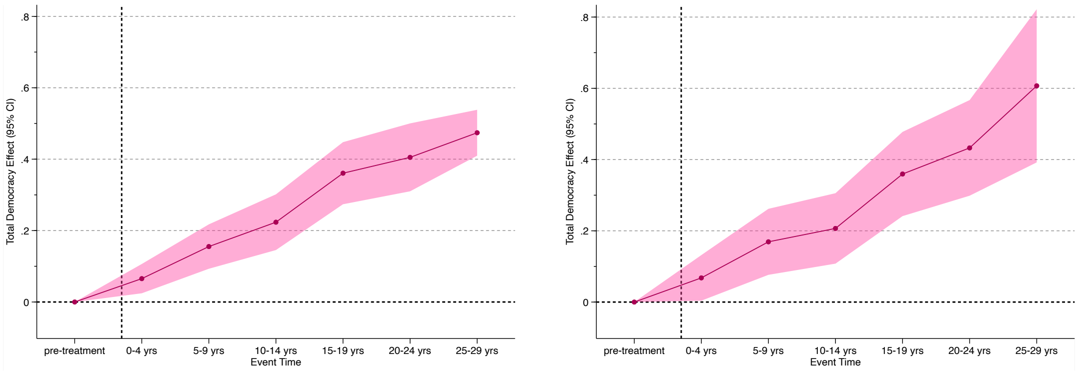
5.4 Dynamic Treatment Effects

Figure 2 presents event-study estimates by 5-year bins relative to treatment onset, for the full sample and the single regime change subsample. The pre-treatment window $[-10, -1]$ is the reference category (normalised to zero). Dummies for the respective 5-year treatment windows are included in our benchmark PPML model with dyadic controls in equation (8).

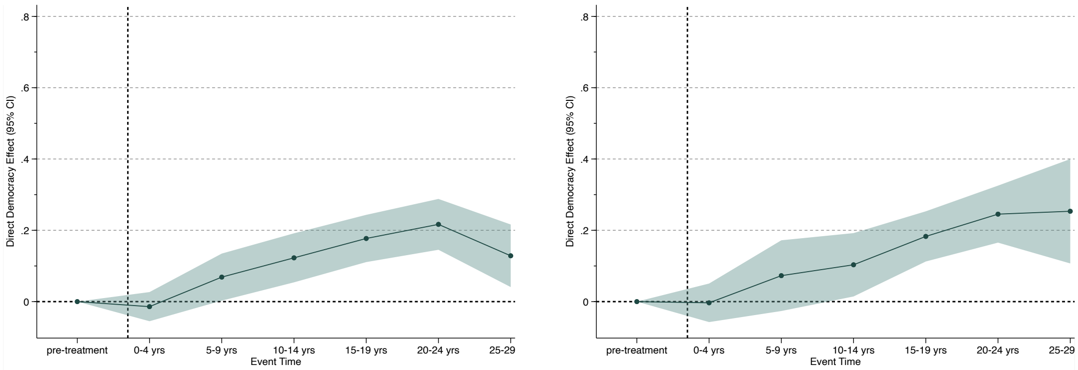
The event-study design serves two purposes. First, the post-treatment path reveals the dynamics of the treatment effect. If democracy operates partly through economic growth, as suggested in the democracy-growth literature [Acemoglu et al., 2019, Boese-Schlosser and Eberhardt, 2024], then the total effect should grow over time as income gains accumulate.

Figure 2: Dynamic treatment effects of democracy on bilateral exports

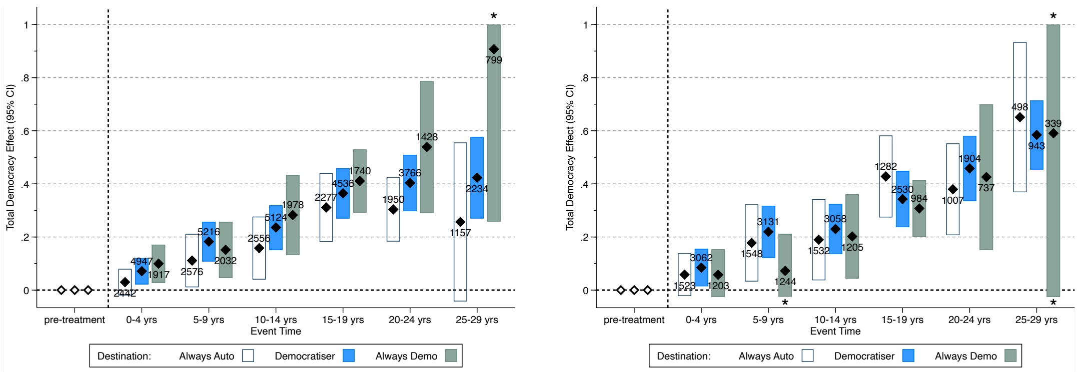
(a) Benchmark — all treated exporters (left), single regime change subsample (right)



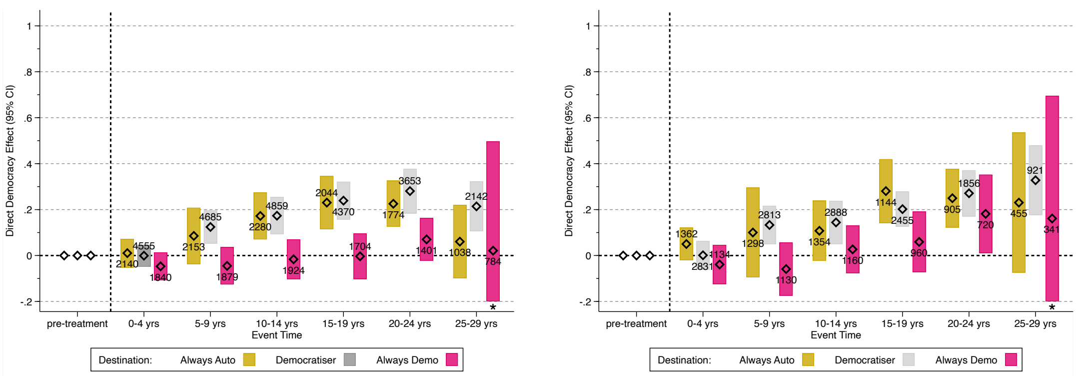
(b) Add GDPpc — all treated exporters (left), single regime change subsample (right)



(c) Benchmark by destination — all treated exporters (left), single regime change subsample (right)



(d) Add GDPpc — all treated exporters (left), single regime change subsample (right)



Notes: Post-treatment event-time bins ($[0,4]$, $[5,9]$, $[10,14]$, $[15,19]$, $[20,24]$, $[25,29]$) relative to treatment onset; 10-year pre-treatment window $[-10,-1]$ normalised to zero. Shaded areas/bars show 95% confidence intervals using two-way clustered SEs. Panels (a)–(b): all pairs, rreg aggregation; panels (c)–(d): stratified by destination regime status, pair count indicated. Left column: full sample; right column: single regime change subsample. $*$ CI curtailed to aid presentation.

Second, stratifying by destination regime status tests for institutional complementarities: if democratic institutions reduce bilateral transaction costs through mutual recognition of contracts and legal standards, effects should be larger for exports to other democracies.

Each bin coefficient is estimated at the pair level from the CCE-DID specification augmented with post-treatment bin dummies: $[0, 4]$, $[5, 9]$, $[10, 14]$, $[15, 19]$, $[20, 24]$, and $[25, 29]$. We restrict the event window to $[-10, 29]$ since data beyond 29 years of treatment $[30, \cdot]$ are too sparse for reliable robust regression aggregation. The omitted category comprises all pre-treatment observations within the event window ($[-10, -1]$), so each coefficient measures the average within-pair change in (log) bilateral exports during that post-treatment window relative to the 10-year pre-treatment baseline (in Appendix Figure C-1 and Table C-2 we use *all* pre-treatment observations as baseline). Not all pairs contribute to every bin: pairs whose exporter democratised later in the sample lack longer post-treatment horizons, and the later bins therefore draw on fewer pairs. We adopt the rreg aggregation practice.

Main Results. The benchmark specification in the left plot of Figure 2 panel (a) reveals a clear post-treatment gradient in the full sample: the rreg coefficient rises from 0.065 over the first five years after democratisation to 0.223 at 10–14 years and 0.361 at 15–19 years, before the gradient eases somewhat to reach 0.474 at 25–29 years. The right plot in the same panel for the single regime-change subsample shows a similar trajectory for early treatment years but but no slowing of growth at longer horizons, reaching 0.607 for 25–29 years. This contrast in magnitudes is consistent with lasting regime change resulting in larger cumulative trade gains and/or attenuation bias from treatment carryover — we explore this further in Section 5.5. The overall growth pattern supports the idea of income accumulation: each additional decade of democratic governance allows further income gains to feed through into higher trade volumes.

The income-channel decomposition in panel (b) confirms this interpretation. When GDP per capita is included, the post-treatment gradient flattens substantially, and the direct effect (net of income) grows more modestly.

Robustness. We use a 10-year pre-treatment window rather than a narrower reference to absorb anticipation effects: economic liberalisation and trade expansion often precede democratic transition [Giavazzi and Tabellini, 2005]. As a robustness check, Appendix Figure C-1 replicates the event study using *all* available pre-treatment data as the reference category. The qualitative pattern is identical, but the effects are larger: the full-sample benchmark rises smoothly from 0.085 at $[0, 4]$ to 0.592 at $[25, 29]$, and the single-regime-change subsample reaches 0.732. The larger magnitudes reflect the longer pre-treatment baseline, which extends further from treatment onset and captures lower trade levels before the anticipation effects that may precede formal transition.

Heterogeneity. Panels (c) and (d) of Figure 2 stratify by destination regime status — Appendix Table C-2 reports the full set of estimates. For the benchmark results in (c), exports to always-democratic destinations show the largest effects, peaking at 0.907 for 25–29 years, followed by exports to fellow democratisers (0.424) and always-autocratic destinations (0.311, peaking in 15-19 years). At longer horizons (25–29 years), effects for always-autocratic destinations dip and become statistically insignificant, pointing to large variation in the treatment effects. If we focus on single regime transformations in the right plot of the same panel, all three sub-groups seem to have similar magnitudes of effects at all treatment length reaching around 0.6 after 25–29 years, with the estimates for always-democratic destinations becoming increasingly imprecise due to small sample size. In panel (d) we can see that increased exports to always-democratic destinations are driven by income effects since the estimates conditioning on income per capita are all insignificant (similarly if we focus on countries with single regime changes). Results for always-autocracies and democratising destinations appear in lock-step, reaching an effect of around 0.2 after 20–24 years before diverging somewhat.

5.5 Diagnostics and Additional Results

Factor spanning test. The CSD diff permutation test for the benchmark specification yields ΔCSD $= -0.004$ with a permutation p-value of 0.623 (999 exporter-level permutations, 86 treated and 50 never-treated control exporters).[12] The null hypothesis of equal residual cross-sectional dependence across groups cannot be rejected at any conventional significance level. The empirical ΔCSD falls squarely within the simulation distribution under $s = 0$ (no confounding), and well below the values associated with economically meaningful bias ($s \geq 0.5$ in Table 3). We conclude that Assumption 1 is supported: control-only CAs provide adequate spanning of the common factor space in this application.

[12] Pairwise correlations are computed on a random subsample of 500 pairs per group to ensure computational tractability; results are stable across subsample draws.

Carryover testing. The geometric persistence model of Section 4.5 parameterises hysteresis through $\rho \in [0,1]$: if $0 < \rho < 1$, trade effects should decay gradually after reversal, producing positive and declining carryover in the early years of autocracy. To test carryover formally, we augment the pair-level PPML with four year-dummies $\gamma_1, \dots, \gamma_4$ for the first four years after each reversal (conditioned on $D_{it} = 0$), alongside a late post-reversal dummy and the standard treatment indicator D_{it}.[13] We run these augmented treatment regressions for trade pairs of the $G = 29$ exporters which experienced reversal to autocracy (or more complex regime dynamics). We compute the Wald test of $H_0\colon \gamma_1 = \gamma_2 = \gamma_3 = \gamma_4 = 0$ as follows: (i) using robust regression we estimate the Mean Group estimate of $\hat{\gamma}_1$ and obtain the applied weights for each exporter pair; (ii) we estimate separate Mean Group regressions for the pairwise estimates of each of the four carryover dummies, adopting the common weights obtained in the previous step; (iii) using seemingly unrelated estimation we compute the Wald statistic for the carryover test across the four Mean Group estimates. The Wald test rejects the null of no carryover in both specifications (benchmark: $p < 0.001$; add GDPpc: $p = 0.035$).

Treatment dynamics for countries with reversal. For countries which experienced autocratic reversal, we need to characterise trade behaviour in each regime spell with respect to the pre-democratic baseline. Using PPML, we estimate for each pair ij of reverter i

$$\mathbb{E}[\text{FLOW}_{ijt} \mid \cdot\,] = \exp\Big\{ \alpha_{ij} + \sum_{k \in \mathcal{K}} \Big[\mu_{ij}^{k} \cdot \text{Spell}_{it}^{k} + \varphi_{ij}^{k} \cdot D_{it}^{0,k} + \vartheta_{ij}^{k} \cdot C_{it}^{k} \cdot \text{Spell}_{it}^{k} \Big] + \delta_{ij} D_{jt} + \boldsymbol{\gamma}_{ij}' \mathbf{X}_{ij,t} + \boldsymbol{\phi}_{ij}' \bar{\mathbf{z}}_{t}^{\text{ctrl}} \Big\} \tag{28}$$

with $\mathcal{K} = \{$1st Dem spell, 1st Reversal, 2nd+ Dem spell, 2nd+ Reversal$\}$ the different spells exporters experienced in and out of democracy following the pre-treatment period.[14] The exporter-time specific terms are defined as follows: Spell_{it}^{k} equals 1 when exporter i is in spell k in year t; $D_{it}^{0,k}$ is an indicator for the first year in spell k (year 0); and C_{it}^{k} counts the cumulative years exporter i has spent in spell k up to year t (zero in year 0, then 1, 2,... and zero in all years not in spell k). $\mathbf{X}_{ij,t}$ represents dyadic controls, D_{jt} the destination regime change dummy and $\bar{\mathbf{z}}_{t}^{\text{ctrl}}$ the CAs for (log) GDP and population computed from control countries.

[13] The late post-reversal dummy absorbs years which are $D{=}0$ but more than four years after the most recent reversal, making pre-treatment (before the first democratisation) the omitted category. The carryover dummies thus compare the reversal year and the three following years against the *pre-treatment* baseline.

[14] We aggregate the effects of later democratic spells or autocratic spells, respectively, and ignore exporter regime change histories — in both cases we otherwise have to deal with substantially reduced sub-samples.

Table 6: Three-Parameter Decomposition — Full Reverter Sample

	Benchmark			+GDPpc		
Regime state k	μ	φ (yr 0)	ϑ (slope)	μ	φ (yr 0)	ϑ (slope)
1st Dem spell	-0.036	-0.105	0.074^{***}	-0.123	-0.079	0.042
	(0.116)	(0.072)	(0.022)	(0.108)	(0.081)	(0.025)
1st Reversal	-0.036	-0.122^{**}	0.069^{**}	-0.154	-0.137^{**}	0.065^{**}
	(0.328)	(0.053)	(0.028)	(0.233)	(0.059)	(0.031)
2nd+ Dem spell	1.270^{***}	-0.337^{***}	0.039^{*}	0.460^{*}	-0.186^{**}	0.046^{***}
	(0.303)	(0.080)	(0.021)	(0.254)	(0.072)	(0.014)
2nd+ Reversal	0.739^{**}	-0.330^{***}	0.039	0.315	-0.245^{**}	0.020
	(0.263)	(0.094)	(0.040)	(0.211)	(0.083)	(0.034)
G (levels)	29	26	26	29	26	26
Pairs		4,236			4,236	

Notes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Reference: pre-democratic period. CGM two-way clustered SEs in parentheses. Predicted log-trade at year 0 of state k: $\mu_k + \varphi_k$; at year $t \geq 1$: $\mu_k + \vartheta_k t$. G indicates the number of exporters after rreg trimming; zero coefficients (dropped dummies) excluded.

For each spell k we distinguish: (i) μ^k, the linear-trend intercept relative to pre-treatment; (ii) φ^k, the first-year deviation from the linear trend; and (iii) ϑ^k, the annualised slope during the spell.[15] Predicted log-trade relative to pre-treatment is $\mu^k + \varphi^k$ in year 0 and $\mu^k + \vartheta^k \cdot t$ in year $t \geq 1$. Results are reported in Table 6. We focus on the benchmark results and use those for the income-augmented specifications to comment on direct vs. indirect effects.

First democratic spell. Exporters enter this spell at the pre-democratic baseline ($\mu = -0.036$, insign.) and experience no significant first-year shock ($\varphi = -0.105$, insign.). Trade then grows at $\exp(0.074) - 1 \approx 7.8\%$ per annum. Once the model includes income, this annual effect becomes statistically insignificant ($\vartheta = 0.042$), hence trade growth during this first spell operates entirely through income.

First reversal. Here, two things happen simultaneously and must be interpreted together: First, $\mu = -0.036$ (insign.) tells us that the linear trend during the first reversal spell has its intercept back at the pre-democratic baseline. This implies a sharp break from the end of the first democratic spell, where accumulated growth at 7.4% per annum over a 5-8 year spell had placed trade roughly 35-60% above pre-treatment. Second, $\varphi = -0.122^{**}$ captures an *additional* first-year deviation. The total drop implied from the end of the first democratic spell

[15] Existing research highlights that democratic regime change is often associated with a (negative) shock to income in the short run, for instance due to a violent regime transition [Cervellati and Sunde, 2014, Acemoglu et al., 2019]. Reversal to autocracy, possibly via a coup, is also easily associated with a negative income shock [Boese-Schlosser and Eberhardt, 2025].

to year 0 of the first reversal is therefore $\varphi + \vartheta \cdot T \approx 40\text{-}60\%$ for typical spell lengths, not just $\varphi \approx 11\%$ $(\exp(-0.122) - 1)$. Crucially, trade *continues to grow* during the first reversal spell, specifically at $\exp(+0.069) - 1 \approx 7.1\%$ per annum. Averaged over the first reversal spell, trade remains at roughly the same level as during the first democratic spell. This implies that there is substantial carryover, in the sense that the average post-reversal trade premium is comparable to the pre-reversal one. The augmented specification indicates that these effects are driven by direct trade mechanisms rather than income ($\varphi = -0.137^{**}$, $\delta = 0.065^{**}$).

Second or later democratic spells. Exporters that democratise for a second (or further) time enter having accumulated substantial trade growth over the preceding spells. This is reflected in the large and significant intercept $\mu = 1.270^{***}$, corresponding to 256% above pre-treatment. This seems very large, but is consistent with a stacking of the gains from the first spell in democracy ($+0.37$–0.59 log points over 5–8 years) and during the first reversal ($+0.35$–0.55 log points over 5–8 years), plus any additional growth at the margin.[16] Entry is then marked by a substantial negative shock of $\exp(-0.337) - 1 \approx -29\%$, while within-spell growth continues at $\exp(0.039) - 1 \approx 4.0\%$ per annum, about half the rate of the first democratic spell. With income included, μ falls sharply, so roughly two-thirds of the elevated level operates through income; the within-spell slope is unchanged, reflecting a direct trade effect.

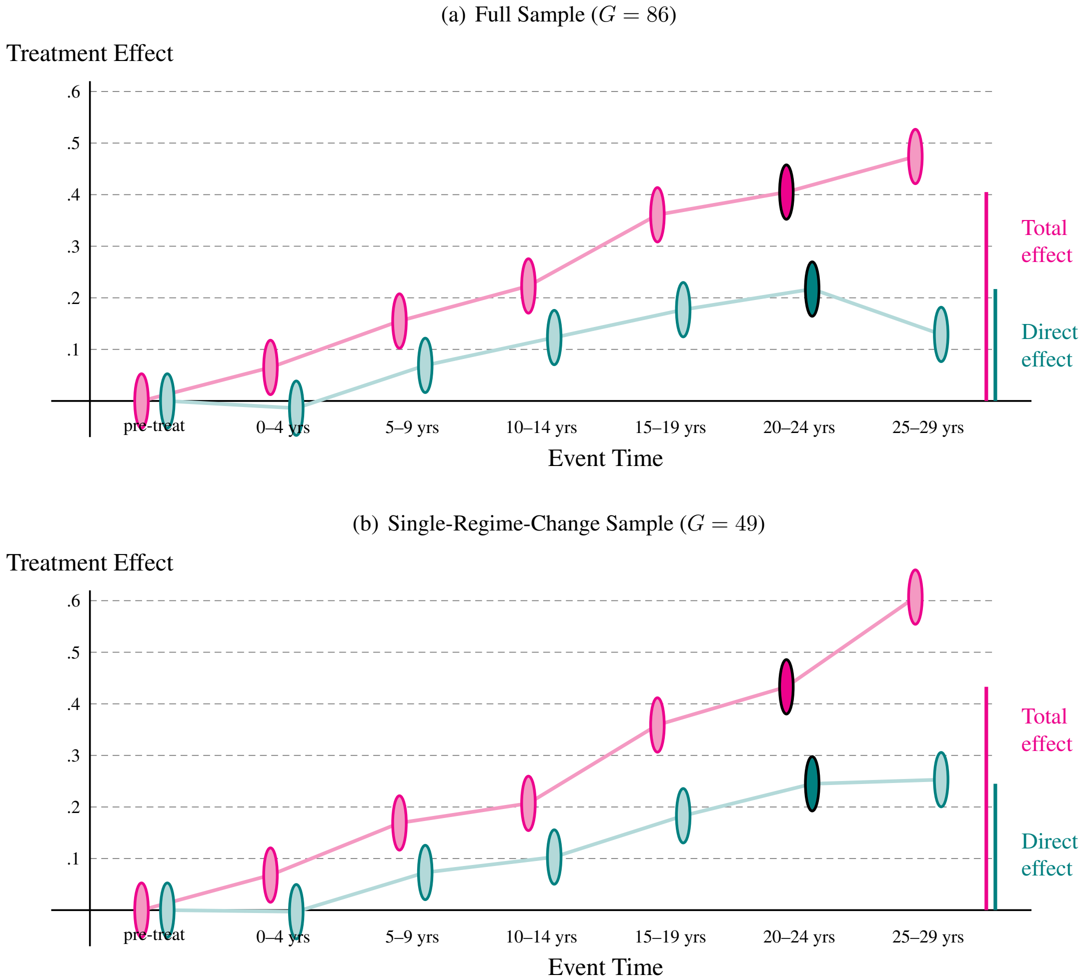
Second or later reversals. Exporters with further reversals enter at $\mu = 0.739^{**}$, substantially lower than for the second and later democratic spells (1.270), given that reversals follow autocratic shocks rather than continuous democratic growth. They suffer a similar first-year shock ($\varphi = -0.330^{***}$, about -28%) and trade is essentially flat thereafter ($\delta = 0.039$, insignificant): carryover is much weaker for these later reversals.

Summary Taken together, these results are consistent with the full-sample patterns in Figure 2(a)–(b). Sharp inter-spell level drops at reversals, combined with lower annual growth in later democratic spells, attenuate the positive carryover observed during the first reversal spell. They ultimately dampen the long-horizon treatment effects in panel (a) relative to the absorbing-treatment benchmark for the single regime change sub-sample in panel (b).

[16]The specification does *not* impose continuity between spells (each spell's μ is estimated independently), so the value of $\mu^{\text{2nd+ Dem}}$ reflects the average trade level observed during these spells, conditional on the pair fixed effect and the CAs, rather than a mechanical sum of prior slopes. The small number of exporters reaching this stage ($G = 20$) also adds noise.

Total vs. direct effects. How important is the 'indirect' income effect for our findings? Studying the gap between the benchmark ATET (0.411) and the income-augmented ATET (0.105) for the full sample is misguided for two reasons: this computation will be distorted by (i) averaging over countries with few and many years in democracy, as well as (ii) any bias introduced by carryover effects.

Figure 3: Treatment Effects — Total vs Direct Effect



Notes: We plot the robust mean estimates by 5-year treatment bin from the specifications including ('direct effect') and excluding ('total effect') GDP per capita for all treated exporters ($G = 86$) in panel (a) and single-regime-change exporters ($G = 49$) in panel (b). The estimates highlighted (magnitudes of treatment effects illustrated with the bars on the right for 20-24 years of treatment) compares the magnitudes for the total and direct treatment effects. We ignore confidence intervals for ease of presentation. The direct effect includes equilibrium adjustment.

Addressing the first point, when we consider treatment effects *by treatment length*, say, after 20-24 years, the estimated ATETs are 0.405 and 0.217, respectively, suggesting a contribution of 0.188 for the income channel (the gap is similar for 15-19 years) — see Figure 3(a).

How distorted are these proportions by the second point, any bias from carryover effects? It turns out not in any significant way: for *single-regime-change exporters* in panel (b) of the same Figure, the total effect amounts to 0.433 after 20-24 years, the direct effect to 0.245, so that the income channel contribution is 0.187 (again, the gaps for 15-19 years are very similar). Hence, we conclude that *roughly half* of the trade gains from democratic regime change are due to the income channel.

This decomposition speaks to two strands of the literature. On the income side, Acemoglu et al. [2019, 47] conclude that democracy raises income "by about 20% in the long run". A 20% income gain, applied through a gravity elasticity of unity, predicts an income-channel contribution of $\ln(1.2) \approx 0.182$.[17] The observed gap in our estimates for 20-24 (15-19) treatment years of 0.188 (0.177) is effectively identical. On the cost side, the direct effect includes any general equilibrium effects: as the income of a democratising country (i) rises, its trading partners' multilateral resistance adjusts, further boosting bilateral flows beyond the direct gravity prediction. This effect may however be moderated by the income-boosting effect of democratic regime change *in other countries* ($-i$). Unlike with the machinery of a structural gravity model (albeit *exclusively* for dyadic treatments and holding all else equal), our methodology cannot isolate the pure trade cost element from that of general equilibrium adjustment for our monadic treatment effect estimate of 0.217 (0.245). Put differently, we can state that the direct, non-income effect of democratic regime change amounts to on average 24% (28%) higher exports after 20-24 (15-19) years.

6 Concluding Remarks

A large class of policy-relevant questions asks how a country-level treatment affects its bilateral flows with all partners: does democratisation expand trade, do capital controls redirect investment, does visa liberalisation alter migration patterns? These questions have remained difficult to answer credibly because the standard econometric toolkit for bilateral flows cannot identify monadic treatment effects, and the recent het-DID revolution has not addressed the dyadic setting. Our paper fills this gap.

The CCE-DID estimator resolves the identification challenges that arise when a monadic

[17] In this context, it needs to be acknowledged (again) that the theoretical elasticity of bilateral trade with respect to GDP is 1 *in the cross-section*, but the effective within-pair elasticity in a panel can differ over points in time.

treatment meets dyadic data: collinearity with directional-time fixed effects, mediator bias from network-wide equilibrium propagation, and heterogeneous pair-level shock exposure. Cross-sectional averages of fundamental variables from never-treated countries ‘replace’ directional-time fixed effects, pair-specific coefficients absorb the heterogeneous factor loadings that the network structure generates, and Mean Group aggregation recovers the ATET without negative weighting. The identifying assumption, factor spanning, is weaker than parallel trends and directly testable through the CSD diff diagnostic.

Our application demonstrates the estimator’s value. Democracy raises bilateral exports by approximately 50%, a large and precisely estimated effect that existing methods cannot recover. The channel decomposition reveals that half of this effect operates through the income gains that democracy generates. This quantifies and connects two previously separate literatures, on democracy and growth and on democracy and trade, within a unified causal framework.

Several directions for future work present themselves: extension to continuous treatments, adaptation to hybrid monadic-dyadic treatment structures (such as unilateral sanctions), formal asymptotic theory for the CCE-DID estimator in the dyadic setting, and structural interpretation of the reduced-form estimates through sufficient statistics. The requirement of a sufficient pool of never-treated countries is a practical limitation; in applications where few countries remain untreated, alternative approaches using not-yet-treated countries with appropriate bias corrections warrant closer investigation.

The CCE-DID estimator is broadly applicable to any setting where a country-level binary treatment affects bilateral flows. By providing both the estimator and accompanying diagnostics, we offer a practical toolkit that enables credible causal inference for monadic treatments in dyadic data.

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Online Appendix – Not Intended for Publication

A Asymptotic Theory — Outline

This appendix provides a sketch of the proofs for consistency and asymptotic normality of the proposed CCE-DID estimator for the average treatment effect on the treated (ATET) in dyadic flow data with heterogeneous factor loadings. We extend the results of Pesaran [2006] and Chan and Kwok [2022] from the linear to the exponential conditional mean (Poisson pseudo-maximum likelihood) setting, and from standard two-way (unit-time) panels to dyadic panels where the cross-sectional dimension indexes country pairs. The key technical challenges are (i) extending the CCE projection argument to the nonlinear PPML setting, which we resolve by exploiting the linearity of the PPML score in $Y_{ij,t}$ and the linearity of the cross-sectional averages in the exponential index, and (ii) the dyadic dependence structure, which requires that inference account for the correlation among pairs sharing a common exporter or importer. We state sufficient conditions and provide proof sketches.

A.1 Setup and Notation

We observe bilateral flows $Y_{ij,t} \geq 0$ for exporter i, importer j, and time $t = 1, \ldots, T$, with N countries acting as both exporters and importers, and treatment indicator $D_{it} \in \{0, 1\}$ varying at the exporter-time level. Let $\mathcal{P}_i = \{(i, j) : j \neq i\}$ denote the set of import partners for exporter i, with $|\mathcal{P}_i| = J_i$, where $| \cdot |$ refers to the cardinality. The total number of pairs is $P = \sum_i J_i$. Treatment is staggered: exporter i first receives treatment at time $E_i \in \{1, \ldots, T\} \cup \{\infty\}$, and may subsequently revert. Treatment timing (and that of possible reversals) will differ across exporters. Not all exporters are treated, in fact a sizeable share N_c is required to remain 'never-treated'. For inference, the number of treated countries N_{tr} will be important, which we also refer to as G.

For each pair (i, j), let $\mathcal{F}_{ij,t}$ denote the information available at time t, comprising the history of treatment indicators $\{D_{is}, D_{js}\}_{s \leq t}$, dyadic controls $\{\mathbf{x}_{ij,s}\}_{s \leq t}$, and cross-sectional averages $\{\bar{\mathbf{z}}_s^{\text{ctrl}}\}_{s \leq t}$. The conditional mean is specified as

$$\mathbb{E}[Y_{ij,t} \mid \mathcal{F}_{ij,t}] = \exp\big(\alpha_{ij} + \theta_{ij}\, D_{it} + \delta_{ij}\, D_{jt} + \mathbf{x}_{ij,t}'\boldsymbol{\beta}_{ij} + \boldsymbol{\phi}_{ij}'\, \bar{\mathbf{z}}_t^{\text{ctrl}}\big), \tag{29}$$

where α_{ij} is a pair fixed effect, θ_{ij} is the pair-specific exporter treatment effect, δ_{ij} captures importer treatment, $\mathbf{x}_{ij,t}$ is a vector of dyadic controls, and $\bar{\mathbf{z}}_t^{\text{ctrl}}$ is the $M \times 1$ vector of cross-sectional averages (CAs) of 'fundamental' variables computed from N_c never-treated control countries.

We collect all pair-specific slope parameters as $\boldsymbol{\psi}_{ij} = (\theta_{ij}, \delta_{ij}, \boldsymbol{\beta}_{ij}', \boldsymbol{\phi}_{ij}')'$, a $(2+K_x+M)\times 1$

vector. The ATET of interest is

$$\theta^{\text{ATET}} = \frac{1}{|\mathcal{M}|} \sum_{(i,j)\in\mathcal{M}} \theta_{ij}, \tag{30}$$

where $\mathcal{M}$ is the set of treated pairs (exporters with $E_i < \infty$) and $|\mathcal{M}|$ is the number of treated pairs.

The aggregation of pair-level estimates $\hat{\theta}_{ij}$ into a single ATET estimate uses robust regression (rreg). Specifically, we regress the $|\mathcal{M}|$ estimated $\hat{\theta}_{ij}$ on a constant using iteratively reweighted least squares with Tukey's biweight function:

$$\hat{\theta}^{\text{rreg}} = \frac{\sum_{(i,j)\in\mathcal{M}} w_{ij}\, \hat{\theta}_{ij}}{\sum_{(i,j)\in\mathcal{M}} w_{ij}}, \tag{31}$$

where $w_{ij} = w(\hat{\theta}_{ij}; \hat{\theta}^{(k)}, \hat{s}^{(k)})$ are the biweight kernel weights at convergence, with $\hat{\theta}^{(k)}$ and $\hat{s}^{(k)}$ denoting the location and scale estimates at the k-th iteration. The rreg procedure begins with an initial Huber M-estimator and iterates until convergence. This estimator downweights pair-level estimates that are outliers — either because of poor pair-level identification or because of genuine extreme heterogeneity — without requiring the researcher to specify trimming thresholds.

A.2 Regularity Conditions

We group assumptions into three blocks: (A) the factor structure and CA requirements, (B) the PPML estimation requirements, and (C) the aggregation requirements.

Block A: Factor Structure and Spanning

Assumption 1 (Factor Structure)**.** *The potential untreated outcome satisfies*

$$Y_{ij,t}(0) = \exp\big(\alpha_{ij} + \boldsymbol{\lambda}'_{ij}\mathbf{F}_t\big) \cdot \varepsilon_{ij,t},$$

where $\mathbf{F}_t$ is a $K \times 1$ vector of common factors, $\boldsymbol{\lambda}_{ij}$ is a pair-specific $K \times 1$ loading vector, and $\eta_{ij,t} > 0$ is a multiplicative idiosyncratic error with $\mathbb{E}[\eta_{ij,t} \mid \mathbf{F}_t, D_{it}, D_{jt}] = 1$ and $\sup_{ij,t} \mathbb{E}[\eta_{ij,t}^{2+\nu}] < \infty$ for some $\nu > 0$. The factors satisfy $T^{-1}\sum_t \mathbf{F}_t\mathbf{F}'_t \xrightarrow{p} \boldsymbol{\Sigma}_F > 0$ as $T \to \infty$. This multiplicative specification is consistent with the exponential conditional mean (29): the condition $\mathbb{E}[\eta_{ij,t} \mid \cdot] = 1$ ensures that $\mathbb{E}[Y_{ij,t}(0) \mid \mathbf{F}_t] = \exp(\alpha_{ij} + \boldsymbol{\lambda}'_{ij}\mathbf{F}_t)$, avoiding the Jensen's inequality problem that would arise from specifying the factor structure in logs.

Assumption 2 (Factor Spanning)**.** *Let $\bar{\gamma}^{z_m}_{ctrl} = N_c^{-1}\sum_{c\in\mathcal{C}} \boldsymbol{\gamma}^{z_m}_c$ denote the average factor loading of the m-th CA observable across N_c never-treated countries $c \in \mathcal{C}$. The $M \times K$ matrix $\bar{\boldsymbol{\Gamma}} = [\bar{\gamma}^{z_1}_{ctrl}, \ldots, \bar{\gamma}^{z_M}_{ctrl}]'$ satisfies $\text{rank}(\bar{\boldsymbol{\Gamma}}) = K$, with $M \geq K$.*

Assumption 3 (CA Approximation)**.** *The idiosyncratic components of the CA observables satisfy a law of large numbers across control countries: as $N_c \to \infty$,*

$$\bar{\mathbf{z}}_t^{ctrl} = \bar{\boldsymbol{\alpha}}_{ctrl} + \bar{\boldsymbol{\Gamma}}\,\mathbf{F}_t + \bar{\mathbf{u}}_t^{ctrl}, \qquad \text{with } \|\bar{\mathbf{u}}_t^{ctrl}\| = O_p(N_c^{-1/2}).$$

Assumptions 1–3 are the dyadic analogues of Assumptions 1–3 in Pesaran [2006], adapted to the setting where $\boldsymbol{\lambda}_{ij}$ is a *pair-level* loading and the CAs are constructed from country-level fundamentals rather than from all pair-level observables (including outcomes). In models of dyadic flows with multilateral resistance, pair-level loadings decompose as

$$\boldsymbol{\lambda}_{ij} = (\boldsymbol{\gamma}_i + \boldsymbol{\gamma}_j) - (\boldsymbol{\pi}_i^F + \boldsymbol{\pi}_j^F),$$

where $\boldsymbol{\gamma}_i$ is a $K \times 1$ vector capturing country i's structural exposure to the common factors — reflecting, for instance, how i's GDP and population respond to global technology and demand shocks (formally, $\boldsymbol{\gamma}_i$ is the country-level analogue of $\boldsymbol{\lambda}_{ij}$, and the observable-specific loadings $\boldsymbol{\gamma}_c^{z_m}$ in Assumption 1 are determined by it) — and $\boldsymbol{\pi}_i^F = \partial \Pi_{it}/\partial \mathbf{F}_t$ is the equilibrium response of i's multilateral resistance term (MRT) to those factors. Similarly for importer j.

Under a CES demand structure, MRT loadings are trade-share-weighted averages of fundamental loadings: $\boldsymbol{\pi}_i^F = \sum_c s_{ic}\,\boldsymbol{\gamma}_c + O(1/N)$. Hence $\boldsymbol{\lambda}_{ij}$ lies in the span of $\{\boldsymbol{\gamma}_c\}_{c=1}^N$. The rank condition in Assumption 1 requires that the control-country CAs span all K dimensions of the common factor space. Since $\bar{\boldsymbol{\Gamma}}$ is $M \times K$ with rank K, its row space is all of $\mathbb{R}^K$: for *any* loading $\boldsymbol{\lambda}_{ij} \in \mathbb{R}^K$ — including those of treated pairs involving treated-country fundamentals $\boldsymbol{\gamma}_i$ — there exist pair-specific CA coefficients $\boldsymbol{\phi}_{ij} \in \mathbb{R}^M$ such that $\boldsymbol{\phi}_{ij}'\bar{\boldsymbol{\Gamma}} = \boldsymbol{\lambda}_{ij}'$ (Lemma 1 below gives the explicit form). In other words, spanning the K-dimensional factor space is sufficient; the control-country CAs do not need to replicate the factor structure of any particular treated country. The MRT loadings, being weighted averages of country fundamentals, inherit this spanning property. This is why country-level CAs work in pair-level regressions with pair-specific coefficients: each pair (i,j) selects the linear combination of CAs that best matches its own loading $\boldsymbol{\lambda}_{ij}$.

Block B: PPML Estimation

Assumption 4 (Conditional Mean)**.** *For each pair (i,j), the conditional mean (29) is correctly specified. The true parameter ψ_{ij}^0 is an interior point of a compact set $\Psi \subset \mathbb{R}^p$.*

Assumption 5 (Pair-Level Identification)**.** *For each pair (i,j), let*

$$\mathbf{w}_{ij,t} = (D_{it},\ D_{jt},\ \mathbf{x}_{ij,t}',\ \bar{\mathbf{z}}_t^{ctrl\prime})'$$

and define the PPML-weighted time-demeaned regressor

$$\tilde{\mathbf{w}}_{ij,t} = \mathbf{w}_{ij,t} - \frac{\sum_{s=1}^{T} \mu^0_{ij,s}\, \mathbf{w}_{ij,s}}{\sum_{s=1}^{T} \mu^0_{ij,s}},$$

where $\mu^0_{ij,t} = \exp(\alpha^0_{ij} + \mathbf{w}'_{ij,t}\boldsymbol{\psi}^0_{ij})$. *Then, as* $T \to \infty$*:*

$$\mathbf{H}_{ij} \equiv_{T\to\infty} \frac{1}{T}\sum_{t=1}^{T} \mu^0_{ij,t}\, \tilde{\mathbf{w}}_{ij,t}\tilde{\mathbf{w}}'_{ij,t} > 0.$$

This requires that, conditional on absorbing pair-specific means, D_{it} retains sufficient time variation that is not collinear with the CA-based factor proxies. In particular, treatment must be staggered: (a) adoption dates E_i must be sufficiently dispersed across treated exporters, since simultaneous adoption by all treated exporters would make D_{it} perfectly collinear with a step-function common factor, violating the rank condition. In addition, (b) each treated exporter must spend a non-negligible fraction of the sample in both the treated and untreated states, i.e., $\min(T^1_i, T^0_i)/T \geq m > 0$ for some m, where $T^1_i = \sum_t D_{it}$ and $T^0_i = T - T^1_i$ are the number of periods spent treated and untreated (allowing for reversals). In practice, exporters with very short pre- or post-treatment windows contribute little to the information matrix and are downweighted by the rreg aggregation.

Assumption 6 (Moment Conditions)**.** *(i)* $\sup_{ij} \mathbb{E}[Y^{2+\nu}_{ij,t}] < \infty$ *for some* $\nu > 0$*. (ii) The PPML score* $\mathbf{s}_{ij,t}(\boldsymbol{\psi}) = (Y_{ij,t} - \mu_{ij,t}(\boldsymbol{\psi}))\, \mathbf{w}_{ij,t}$ *satisfies a functional CLT:*

$$\frac{1}{\sqrt{T}}\sum_{t=1}^{T} \mathbf{s}_{ij,t}(\boldsymbol{\psi}^0_{ij}) \xrightarrow{d} \mathcal{N}(\mathbf{0}, \mathbf{V}_{ij}),$$

where $\mathbf{V}_{ij}$ *is the long-run variance of the score, allowing for serial dependence in* $(Y_{ij,t}, \mathbf{w}_{ij,t})$*. (iii) The conditional variance satisfies*

$$\mathit{Var}(Y_{ij,t} \mid \mathcal{F}_{ij,t}) = \sigma^2_{ij}\, \mu^{\kappa}_{ij,t}$$

for some $\kappa \in [1, 2]$ *and* $\sigma^2_{ij} > 0$*, uniformly over* ij *and* t*. (iv) The exponential index is uniformly bounded with probability approaching one:* $\sup_{ij,t} |\mathbf{w}'_{ij,t}\boldsymbol{\psi}^0_{ij}| \leq C$ *w.p.a.1 for some* $C < \infty$*, where the probability is over the CA approximation error in* $\bar{\mathbf{z}}^{ctrl}_t$*.*

This ensures that the CA approximation error $r_{ij,t} = O_p(N_c^{-1/2})$ in the index propagates through $\exp(\cdot)$ with bounded derivatives: $|\exp(a + r) - \exp(a)| \leq \exp(C)\,|r|\,e^{|r|}$, which is $O_p(N_c^{-1/2})$ since $e^{|r|} \to 1$ when $r \to 0$. The key implication is that the nonlinear transformation does *not* amplify the error: an $O_p(N_c^{-1/2})$ error in the index remains $O_p(N_c^{-1/2})$ in the conditional mean, rather than being inflated to a larger order.

Block C: Robust Regression Aggregation

Assumption 7 (Random Coefficients)**.** *The pair-specific treatment effects satisfy*

$$\theta_{ij} = \theta^{ATET} + \upsilon_{ij}, \qquad \mathbb{E}[\upsilon_{ij}] = 0, \quad \mathbb{E}[\upsilon_{ij}^2] = \sigma_\upsilon^2 < \infty.$$

The distribution of υ_{ij} is symmetric about zero. The estimation errors $\hat{\theta}_{ij} - \theta_{ij}$ are mean-zero conditional on the treatment effects.

Assumption 8 (Two-Way Cluster Dependence)**.** *Let $\mathcal{M}_i = \{(i, j) : (i, j) \in \mathcal{M}\}$ and $\mathcal{M}_{\cdot j} = \{(i', j) : (i', j) \in \mathcal{M}\}$ denote the sets of treated pairs sharing exporter i and importer j, respectively, with G treated exporters in total. The pair-level estimation errors $(\hat{\theta}_{ij} - \theta_{ij})$ exhibit two sources of within-sample dependence:*

- Exporter clustering: *pairs sharing exporter i are correlated because they share D_{it} and exporter-level shocks.*
- Importer clustering: *pairs sharing importer j are correlated because they share D_{jt} and importer-level shocks.*

Conditional on $\mathbf{F} = (\mathbf{F}_1, \ldots, \mathbf{F}_T)'$, pairs that share neither exporter nor importer are approximately independent: their residual dependence operates only through equilibrium spillovers, which are of order $O(N^{-1})$ and satisfy α-mixing with coefficients declining at rate $O(N^{-1})$, ensuring that they do not affect the CLT [see Graham, 2020]. The two-way-clustered estimation errors satisfy

$$\frac{1}{\sqrt{G}} \sum_{i=1}^{G} \bar{e}_i \xrightarrow{d} \mathcal{N}(0, \sigma_e^2) \qquad \text{as } G \to \infty,$$

where $\bar{e}_i = |\mathcal{M}_i|^{-1} \sum_{(i,j)\in\mathcal{M}_i} (\hat{\theta}_{ij} - \theta_{ij})$ and σ_e^2 incorporates both within-exporter and within-importer correlation. The number of partners $J_i = |\mathcal{M}_i|$ satisfies $\sup_i J_i/\bar{J} < \infty$, where $\bar{J} = G^{-1} \sum_i J_i$, so that no single exporter dominates the aggregate.

Remark 1 (Effective Cross-Section)**.** *Although the rreg estimator (31) operates on $|\mathcal{M}|$ pair-level estimates, the effective cross-section for inference is G (treated exporters), not $|\mathcal{M}|$ (treated pairs). Pairs sharing an exporter are correlated through D_{it} and exporter-level shocks; pairs sharing an importer are correlated through D_{jt} and importer-level shocks. Two-way clustering by exporter and importer [Cameron et al., 2011] accounts for both channels of dependence. Crucially, importer-side clustering does not further reduce the effective dimension below G: the two-way clustered variance decomposes as $\hat{V}_{exp} + \hat{V}_{imp} - \hat{V}_{pair}$ (Corollary 1), which inflates the variance relative to exporter-only clustering but does not change the number of independent units driving the $\sqrt{G}$ convergence rate. The rate of convergence is therefore $\sqrt{G}$, requiring $G \to \infty$ and $\sqrt{G}/T \to 0$.*

A.3 Main Results

Lemma 1 (CA Factor Proxy)**.** *Under Assumptions 1–3, as* $N_c \to \infty$*,*

$$\bar{\mathbf{z}}_t^{ctrl} = \bar{\boldsymbol{\alpha}}_{ctrl} + \bar{\boldsymbol{\Gamma}}\,\mathbf{F}_t + O_p(N_c^{-1/2}),$$

and since $\operatorname{rank}(\bar{\boldsymbol{\Gamma}}) = K$*, there exists a* $K \times M$ *matrix* $\mathbf{A}$ *such that* $\mathbf{F}_t = \mathbf{A}\,(\bar{\mathbf{z}}_t^{ctrl} - \bar{\boldsymbol{\alpha}}_{ctrl}) + O_p(N_c^{-1/2})$*. Therefore, including* $\bar{\mathbf{z}}_t^{ctrl}$ *with pair-specific coefficients* $\boldsymbol{\phi}_{ij}$ *is equivalent to including* $\mathbf{F}_t$ *with pair-specific loadings* $\boldsymbol{\lambda}_{ij}$*, where the CA coefficients satisfy* $\boldsymbol{\phi}_{ij} = \bar{\boldsymbol{\Gamma}}\,(\bar{\boldsymbol{\Gamma}}'\bar{\boldsymbol{\Gamma}})^{-1}\boldsymbol{\lambda}_{ij}$*, up to a rotation, with an approximation error of order* $O_p(N_c^{-1/2})$*.*

Proof sketch. The result follows directly from Assumption 3 and the rank condition in Assumption 1. Write $\bar{\mathbf{z}}_t^{\text{ctrl}} - \bar{\boldsymbol{\alpha}}_{\text{ctrl}} = \bar{\boldsymbol{\Gamma}}\,\mathbf{F}_t + \bar{\mathbf{u}}_t^{\text{ctrl}}$. Left-multiplying by $(\bar{\boldsymbol{\Gamma}}'\bar{\boldsymbol{\Gamma}})^{-1}\bar{\boldsymbol{\Gamma}}'$ yields $\mathbf{F}_t = (\bar{\boldsymbol{\Gamma}}'\bar{\boldsymbol{\Gamma}})^{-1}\bar{\boldsymbol{\Gamma}}'(\bar{\mathbf{z}}_t^{\text{ctrl}} - \bar{\boldsymbol{\alpha}}_{\text{ctrl}}) + O_p(N_c^{-1/2})$, since $\bar{\boldsymbol{\Gamma}}$ has full column rank by Assumption 1 and the remainder is $O_p(N_c^{-1/2})$ by Assumption 3. Substituting into the factor structure shows that any pair-specific linear combination $\boldsymbol{\lambda}_{ij}'\mathbf{F}_t$ can be written as $\boldsymbol{\phi}_{ij}'\bar{\mathbf{z}}_t^{\text{ctrl}} + c_{ij} + O_p(N_c^{-1/2})$, where c_{ij} is absorbed by α_{ij}. □

Proposition 2 (Pair-Level Consistency)**.** *Let Assumptions 1–6 hold. For each pair* (i, j)*, the PPML estimator* $\hat{\boldsymbol{\psi}}_{ij}$ *from* (29) *with pair fixed effects* α_{ij} *satisfies*

$$\hat{\boldsymbol{\psi}}_{ij} - \boldsymbol{\psi}_{ij}^0 = O_p(T^{-1/2}) + O(N_c^{-1/2}) \qquad \text{as } T, N_c \to \infty,$$

and, in particular, $\hat{\theta}_{ij} - \theta_{ij} = O_p(T^{-1/2}) + O(N_c^{-1/2})$*.*

Moreover, as $T, N_c \to \infty$ *with* $T/N_c \to 0$ *(so that the CA approximation bias is asymptotically negligible relative to the pair-level sampling variance):*

$$\sqrt{T}\,(\hat{\boldsymbol{\psi}}_{ij} - \boldsymbol{\psi}_{ij}^0) \xrightarrow{d} \mathcal{N}(\mathbf{0},\ \mathbf{H}_{ij}^{-1}\,\mathbf{V}_{ij}\,\mathbf{H}_{ij}^{-1}),$$

where $\mathbf{H}_{ij}$ *and* $\mathbf{V}_{ij}$ *are defined in Assumptions 5 and 6. If* $T/N_c \not\to 0$*, the asymptotic distribution acquires a non-zero mean of order* $\sqrt{T/N_c}$ *reflecting the CA approximation bias. Since all pairs include the same CA variables* $\bar{\mathbf{z}}_t^{ctrl}$*, this bias is common to all pair-level estimates and does not cancel across pairs. However, it is* not *required for the rreg ATET result (Proposition 3), which needs only the weaker condition* $\sqrt{G}/N_c \to 0$*: the CA bias enters the aggregate at order* $O(N_c^{-1/2})$*, and* $\sqrt{G} \cdot O(N_c^{-1/2}) = o(1)$ *under* $\sqrt{G}/N_c \to 0$*, regardless of* T/N_c*.*

Proof sketch. The argument proceeds in two steps.

Step 1: Reduction to a correctly specified model. By Lemma 1, including $\bar{\mathbf{z}}_t^{\text{ctrl}}$ with pair-specific coefficients is equivalent (up to $O_p(N_c^{-1/2})$) to including the true factors $\mathbf{F}_t$ with pair-specific loadings. Substituting into (29), the conditional mean becomes

$$\mathbb{E}[Y_{ij,t} \mid \mathcal{F}_{ij,t}] = \exp\!\big(\alpha_{ij}^* + \theta_{ij}\,D_{it} + \delta_{ij}\,D_{jt} + \mathbf{x}_{ij,t}'\boldsymbol{\beta}_{ij} + \boldsymbol{\lambda}_{ij}'\mathbf{F}_t + r_{ij,t}\big),$$

where α^*_{ij} absorbs the intercept shift from the CA constant and $r_{ij,t} = O_p(N_c^{-1/2})$ uniformly over t. Under Assumption 4, the factor-augmented conditional mean is correctly specified, so the PPML objective function is maximised at the true parameter up to the $O_p(N_c^{-1/2})$ CA approximation error.

Step 2: PPML consistency under correct specification. With the factors (approximately) absorbed, we are in the setting of Gouriéroux et al. [1984]: the PPML estimator is consistent for the parameters of the exponential conditional mean under correct specification, provided the information matrix is nonsingular (Assumption 5) and sufficient moment conditions hold (Assumption 6). The pair-level PPML estimator is obtained by estimating (29) separately for each pair (i, j). The pair fixed effect admits the closed-form solution

$$\hat{\alpha}_{ij}(\boldsymbol{\psi}) = \ln\Big(\sum_t Y_{ij,t} \Big/ \sum_t \exp(\mathbf{w}'_{ij,t}\boldsymbol{\psi})\Big),$$

so the profiled Poisson quasi-log-likelihood for pair (i, j) is

$$\ell_{ij}(\boldsymbol{\psi}_{ij}) = \sum_{t=1}^{T} \Big[Y_{ij,t}\, \mathbf{w}'_{ij,t}\boldsymbol{\psi}_{ij} - \hat{n}_{ij}(\boldsymbol{\psi}_{ij}) \exp(\mathbf{w}'_{ij,t}\boldsymbol{\psi}_{ij})\Big],$$

where $\hat{n}_{ij}(\boldsymbol{\psi}_{ij}) = \sum_t Y_{ij,t} \big/ \sum_t \exp(\mathbf{w}'_{ij,t}\boldsymbol{\psi}_{ij})$ is the concentrated fixed effect. Standard arguments [see, e.g., Newey and McFadden, 1994, Theorem 2.1] yield uniform convergence of $T^{-1}\ell_{ij}$ to its population limit, which is uniquely maximised at $\boldsymbol{\psi}^0_{ij}$ under Assumptions 4–5, giving consistency.

The $O(N_c^{-1/2})$ bias arises because the CAs are noisy proxies for the true factors. This 'generated regressor' bias is of order $N_c^{-1/2}$ and vanishes as $N_c \to \infty$, as shown in Pesaran [2006] for the linear case. The nonlinearity of the exponential link does not affect the order of this bias term because the PPML first-order conditions are linear in $Y_{ij,t}$ (though nonlinear in $\boldsymbol{\psi}_{ij}$), and the generated regressor enters linearly in the index.

Asymptotic normality follows from the standard sandwich formula for M-estimators, with $\mathbf{H}_{ij}$ as the expected Hessian and $\mathbf{V}_{ij}$ as the score variance. The latter may differ from $\mathbf{H}_{ij}$ due to overdispersion (Assumption 6 (iii)), necessitating the robust sandwich form. □

Proposition 3 (Consistency and Normality of the rreg ATET Estimator)**.** *Let Assumptions 1–8 hold. The robust regression estimator* (31) *satisfies:*

$$\hat{\theta}^{rreg} \xrightarrow{p} \theta^{ATET}$$

as $G, T, N_c \to \infty$ *with* $\sqrt{G}/T \to 0$ *and* $\sqrt{G}/N_c \to 0$.

Furthermore, with two-way (exporter × importer) clustered standard errors:

$$\sqrt{G}\,(\hat{\theta}^{rreg} - \theta^{ATET}) \xrightarrow{d} \mathcal{N}(0,\ \sigma^2_{\bar{v}} + \sigma^2_e),$$

where $\sigma^2_{\bar{\upsilon}} = \mathrm{Var}(\bar{\upsilon}_i)$ is the across-exporter variance of the exporter-level average treatment effects (with $\bar{\upsilon}_i = |\mathcal{M}_i|^{-1}\sum_{j\in\mathcal{M}_i}\upsilon_{ij}$) and σ^2_e is the average estimation-error variance incorporating both exporter-level and importer-level correlation from Assumption 8.

Proof sketch. Write $\hat{\theta}_{ij} = \theta^{\text{ATET}} + \upsilon_{ij} + (\hat{\theta}_{ij} - \theta_{ij})$, where υ_{ij} is the random coefficient and $\hat{\theta}_{ij} - \theta_{ij}$ is the estimation error.

The rreg estimator is a weighted average (31). Under Assumption 7, the υ_{ij} are symmetric about zero, so the rreg estimator — being a consistent estimator of the location of a symmetric distribution [Huber, 1981] — targets θ^{ATET}. It remains to verify that the estimation errors do not shift the location.

Decompose $\hat{\theta}^{\text{rreg}} - \theta^{\text{ATET}}$ into three terms. At convergence, the rreg weights w_{ij} depend on $\hat{\theta}_{ij}$ and hence on both υ_{ij} and the estimation error. However, under standard M-estimator asymptotics [Huber, 1981], the converged weights can be treated as evaluated at the population values up to $o_p(1)$ terms, yielding the linearised decomposition:

$$\hat{\theta}^{\text{rreg}} - \theta^{\text{ATET}} = \underbrace{\frac{\sum_{(i,j)} w^*_{ij}\,\upsilon_{ij}}{\sum_{(i,j)} w^*_{ij}}}_{\text{(I): coefficient heterogeneity}} + \underbrace{\frac{\sum_{(i,j)} w^*_{ij}\,(\hat{\theta}_{ij} - \theta_{ij})}{\sum_{(i,j)} w^*_{ij}}}_{\text{(II): estimation error}} + \underbrace{O(N_c^{-1/2})}_{\text{(III): CA bias}} + o_p(|\mathcal{M}|^{-1/2}), \quad (32)$$

where $w^*_{ij} = w(\upsilon_{ij};\, 0,\, s^*)$ are the population-level biweight kernel weights evaluated at the true location (zero, since $\mathbb{E}[\upsilon_{ij}] = 0$) and the population scale $s^* = \mathrm{MAD}(\upsilon_{ij})$ (the median absolute deviation of the pair-level treatment-effect heterogeneity), and the $o_p(|\mathcal{M}|^{-1/2})$ remainder captures the linearisation error from replacing converged weights with population weights.

Term (I). Under Assumption 7, υ_{ij} is symmetric with mean zero and finite variance. At convergence, the rreg weights w_{ij} are bounded, positive, and depend on υ_{ij} only through its magnitude. By the consistency theory for M-estimators with bounded ψ-functions [Huber, 1981], this weighted average converges to the centre of symmetry of υ_{ij}, which is zero. The rate is governed by the number of independent clusters. Under Assumption 8, the effective number of independent units is G (exporters). The exporter-level averages $\bar{\upsilon}_i = |\mathcal{M}_i|^{-1}\sum_{j\in\mathcal{M}_i}\upsilon_{ij}$ satisfy $\mathbb{E}[\bar{\upsilon}_i] = 0$ and $\mathrm{Var}(\bar{\upsilon}_i) = \sigma^2_\upsilon[1 + (J_i - 1)\rho_i]/J_i$, where ρ_i is the within-exporter pairwise correlation of υ_{ij}. Even though the number of partners J_i varies across exporters, the Lindeberg condition is satisfied because $\sup_i \mathbb{E}[\bar{\upsilon}_i^2\,\mathbf{1}(|\bar{\upsilon}_i| > \varepsilon\sqrt{G})] \to 0$ for all $\varepsilon > 0$, given the finite second moments from Assumption 7 and the boundedness of $J_i/\bar{J}$ where $\bar{J} = G^{-1}\sum_i J_i$. Hence $(1/\sqrt{G})\sum_i \bar{\upsilon}_i \xrightarrow{d} \mathcal{N}(0, \sigma^2_{\bar{\upsilon}})$.

Term (II). By Proposition 2, $\hat{\theta}_{ij} - \theta_{ij} = O_p(T^{-1/2})$ for each pair. The rreg weighting does not alter the order: the weighted average of estimation errors converges to zero at rate $O_p((GT)^{-1/2})$, which is $o_p(G^{-1/2})$ under $\sqrt{G}/T \to 0$. At the exporter-cluster level, $\bar{e}_i = O_p(T^{-1/2})$, and by Assumption 8, $(1/\sqrt{G})\sum_i \bar{e}_i \xrightarrow{d} \mathcal{N}(0, \sigma^2_e)$.

Term (III). Each pair-level estimate carries a CA approximation bias of $O(N_c^{-1/2})$ from Proposition 2. The weighted average preserves this order: the aggregate bias is $O(N_c^{-1/2})$. Under $\sqrt{G}/N_c \to 0$, this term is $o(G^{-1/2})$ and negligible in the $\sqrt{G}$-scaled limit.

Combining: $\sqrt{G}(\hat{\theta}^{\text{rreg}} - \theta^{\text{ATET}}) = \sqrt{G} \cdot \text{(I)} + \sqrt{G} \cdot \text{(II)} + o_p(1)$. Since (I) reflects population heterogeneity and (II) reflects sampling variation, the two are asymptotically independent and the variances add. □

Corollary 1 (Consistent Variance Estimation)**.** *Under the conditions of Proposition 3, a consistent estimator of* $\text{Var}(\hat{\theta}^{rreg})$ *is given by the two-way clustered variance [Cameron et al., 2011]:*

$$\widehat{\text{Var}}(\hat{\theta}^{rreg}) = \hat{V}_{exp} + \hat{V}_{imp} - \hat{V}_{pair},$$

where $\hat{V}_{exp}$, $\hat{V}_{imp}$, *and* $\hat{V}_{pair}$ *are the cluster-robust Huber–White variances from the rreg, clustering by exporter, importer, and exporter-importer pair, respectively. Specifically,*

$$\hat{V}_{exp} = \frac{1}{W^2} \sum_{i=1}^{G} \left(\sum_{j \in \mathcal{M}_i} w_{ij} \, (\hat{\theta}_{ij} - \hat{\theta}^{rreg}) \right)^2, \qquad W = \sum_{(i,j) \in \mathcal{M}} w_{ij},$$

and $\hat{V}_{imp}$ *and* $\hat{V}_{pair}$ *are defined analogously, clustering by importer and by pair. This estimator accounts for both the within-exporter correlation (shared* D_{it} *and exporter shocks) and the within-importer correlation (shared* D_{jt} *and importer shocks) described in Assumption 8. It automatically captures both treatment-effect heterogeneity (*$\sigma_{\bar{v}}^2$*) and estimation noise (*σ_e^2*) without requiring separate estimation of either component, in the same way that the cross-sectional variance does in the heterogeneous panel literature [Pesaran, 2006, Remark 3].*

A.4 The Role of Nonlinearity

The extension from linear CCE to CCE-DID is not trivial, as the PPML objective is a nonlinear function of the parameters. We address three specific concerns.

(a) Index linearity and generated regressors. Although the PPML likelihood is nonlinear, the CAs enter *linearly* in the exponential index: $\boldsymbol{\phi}_{ij}' \bar{\mathbf{z}}_t^{\text{ctrl}}$ appears inside the $\exp(\cdot)$ alongside other linear-in-parameters terms. This means the 'generated regressor' analysis of Pesaran [2006] — showing that the CA approximation error has asymptotically negligible effects — applies to the index directly. A natural concern is whether the nonlinear transformation $\exp(\cdot)$ might amplify the $O_p(N_c^{-1/2})$ index error into a larger-order error in the conditional mean. It does not: the index boundedness condition (Assumption 6 (iv)) ensures that

$$|\exp(a + r) - \exp(a)| = \exp(a)\,|e^r - 1| \leq \exp(C)\,|r|\,e^{|r|},$$

so for $r = O_p(N_c^{-1/2})$ small, $|e^r - 1| = |r|(1 + o_p(1))$ and the error in the conditional mean remains $O_p(N_c^{-1/2})$ — the same order as in the index, not a larger one. The key requirement is that the CA approximation error enters the PPML score only through the conditional mean $\mu_{ij,t}$, and the score is linear in $Y_{ij,t}$ (point (b) below), so the generated-regressor bias in the first-order conditions inherits the $O_p(N_c^{-1/2})$ order from the index error without amplification.

(b) PPML first-order conditions. The PPML first-order conditions are

$$\sum_{t=1}^{T}(Y_{ij,t} - \hat{\mu}_{ij,t})\, \mathbf{w}_{ij,t} = \mathbf{0},$$

which are *linear* in the dependent variable $Y_{ij,t}$ (though nonlinear in parameters through $\hat{\mu}_{ij,t}$). This linearity in Y is the key property exploited by Santos Silva and Tenreyro [2006]: it ensures robustness to heteroskedasticity and means that the estimator targets the conditional mean parameters regardless of the true conditional distribution. The same property ensures that the factor-augmentation strategy works: the CAs need only absorb the *mean* of the common component $\exp(\boldsymbol{\lambda}_{ij}'\mathbf{F}_t)$, which they do through the linear index with pair-specific coefficients, just as in the linear case.

(c) Incidental parameters. The pair fixed effects α_{ij} are consistently estimated in the PPML setting as $T \to \infty$, because the Poisson quasi-likelihood does not suffer from the incidental parameter bias that affects logit and probit FE estimators. This holds even in the presence of interactive fixed effects: Chen et al. [2021] prove formally that in nonlinear factor models estimated by fixed effects MLE, the Poisson specification yields zero asymptotic bias for both slope parameters and average partial effects, regardless of the number of factors. The key structural property is that the Poisson log-likelihood satisfies $\partial_z^3 \ell_{ij} = \partial_z^2 \ell_{ij} = -\mu_{ij,t}^0$ (where $\mu_{ij,t}^0 = \exp(z_{ij,t}^0)$ is the conditional mean), which causes all incidental parameter bias terms to vanish (their Theorem 1, Example 3). Since we estimate each pair separately with $T \to \infty$ rather than jointly with $I, J \to \infty$, our setting is simpler than theirs, and the same no-bias property applies *a fortiori*. The concentrated likelihood approach used in estimation is numerically equivalent to conditioning on the sufficient statistic $\sum_t Y_{ij,t}$ for α_{ij}, and the resulting estimator of the slope parameters is $\sqrt{T}$-consistent without bias correction. The rate condition $\sqrt{G}/T \to 0$ already implies $T \to \infty$, which is the condition required for FE consistency; hence no additional rate restriction is needed beyond those stated in Proposition 3.

A.5 Discussion of Key Conditions

Rate condition $\sqrt{G}/T \to 0$

This condition requires that the time dimension grows faster than the square root of the number of treated exporters, ensuring that pair-level estimation errors are small enough to be averaged out by the rreg aggregation. In the empirical application, $G = 86$ (or 49 for single regime change) and $T = 65$, giving $\sqrt{G}/T \approx 0.14$ (or 0.11). These ratios are well below 1, and the accompanying simulation evidence confirms negligible bias at these dimensions.

Rate condition $\sqrt{G}/N_c \to 0$

This condition requires enough control countries for the CA approximation error to be negligible. With $G \leq 86$ and $N_c = 50$, we have $\sqrt{G}/N_c \approx 0.19$. While not vanishing, the simulation evidence shows that the CA approximation bias is small relative to the true ATET, confirming that the asymptotic approximation is adequate for the empirical sample.

Two-way dependence (Assumption 8)

Pair-level estimation errors exhibit two channels of dependence beyond the common factors (which are absorbed by CAs). The first channel is shared country identity: pairs sharing an exporter i are correlated through D_{it} and exporter-level idiosyncratic shocks, while pairs sharing an importer j are correlated through D_{jt} and importer-level shocks. The second channel is equilibrium propagation through the multilateral resistance system. The latter is second-order: after conditioning on $\mathbf{F}$, the residual equilibrium spillover from country i's treatment to a pair (i', j') that shares neither exporter nor importer with i is of order $O(1/N)$, as each country's weight in the global system is approximately $1/N$ (or smaller). This residual dependence satisfies the α-mixing conditions required for the CLT in Assumption 8 [cf. Graham, 2020, on conditionally independent dyads]. The two-way clustering by exporter and importer [Cameron et al., 2011] accounts for the first channel, while the second channel is asymptotically negligible under $N \to \infty$.

Symmetry of υ_{ij} (Assumption 7)

The rreg aggregation targets the centre of symmetry of the distribution of $\hat{\theta}_{ij}$, which equals θ^{ATET} under Assumption 7. If the distribution of υ_{ij} is asymmetric — for instance, if treatment effects were systematically larger for pairs involving small importers — then rreg targets the centre of symmetry rather than the mean. Under asymmetry, the estimand shifts from θ^{ATET} (the mean of θ_{ij}) to the median, which remains a well-defined causal parameter: the median pair-level treatment effect on the treated. For moderate skewness, the mean–median gap is small; the accompanying simulation evidence confirms that the difference is negligible across a range of DGPs with heterogeneous effects. We note that the symmetry assumption is standard in the robust statistics literature for M-estimation [Huber, 1981] and is no more restrictive than the implicit symmetry assumptions underlying trimmed-mean or median-based aggregators used elsewhere in the heterogeneous treatment effects literature.

The spanning condition

A distinctive advantage of the CCE-DID approach is that the key identifying assumption (factor spanning, Assumption 1) is empirically testable. Under correct spanning, the PPML Pearson residuals $\hat{e}_{ij,t} = (Y_{ij,t} - \hat{\mu}_{ij,t})/\hat{\mu}_{ij,t}^{1/2}$ should be free of systematic cross-sectional dependence (CSD). The CSD test statistic [Pesaran, 2015], computed as a standardised average of pairwise residual correlations, satisfies CSD $\xrightarrow{d} \mathcal{N}(0,1)$ under the null of weak cross-sectional dependence in the residuals. The CSD diff diagnostic — the difference in CSD between treated and control subsamples — provides a direct test for the asymmetric loading problem: a significant positive CSD diff indicates that the CAs inadequately proxy the factors relevant to treated pairs. In contrast, the parallel trends assumption underlying standard DID is inherently untestable because it concerns counterfactual outcomes.

A.6 Relationship to Existing Results

Table A-7 situates our results relative to the closest existing asymptotic theory.

Table A-7: Comparison with Existing Asymptotic Results

	P (2006)	CK (2022)	This paper
Model	Linear	Linear DID	Exp. (PPML) DID
Cross-section	Units	Units	Pairs
Treatment	None	Staggered	Staggered
Factor proxies	All CAs	Control CAs	Control CAs
Het. slopes	λ_i	λ_i	$\boldsymbol{\lambda}_{ij}$
GE/network	No	No	Yes
Aggregation	MG	MG	rreg
Conv. rate	$\sqrt{N}$	$\sqrt{N}$	$\sqrt{G}$

P = Pesaran [2006]; CK = Chan and Kwok [2022].

Our contribution is to bridge the gap between Chan and Kwok [2022]'s linear PCDID framework and the nonlinear PPML setting with dyadic dependence. Our approach is complementary to Chen et al. [2021], who develop fixed effects estimation of nonlinear factor models — including Poisson — for panel and network data with interactive unobserved heterogeneity. Their results confirm that dyadic flows naturally admit a rich factor structure capturing multilateral resistance, homophily, and clustering, and that in the Poisson case slope parameters are free of incidental parameter bias even when the number of factors is large. We differ in two key respects: rather than estimating the full factor structure (α_i, γ_j) jointly, we use cross-sectional averages from never-treated controls as CCE proxies, thereby avoiding high-dimensional factor estimation and the associated bias corrections; and we study a different estimand — monadic treatment effects under staggered adoption — with a heterogeneous DID estimator and two-way dyadic clustering.

The key new elements are: (i) the structural decomposition establishing that pair-level loadings in empirical models of dyadic flow data are spanned by country-level fundamental loadings, justifying the use of country-level CAs in pair-level regressions; (ii) the observation that the PPML index linearity in CAs preserves, and does not amplify, the order of the CA approximation bias from the linear case; (iii) the recognition that the effective cross-section for aggregation is G (treated exporters), not $|\mathcal{M}|$ (treated pairs), with two-way clustering by exporter and importer to account for within-country dependence; and (iv) the use of robust regression rather than simple averaging, which downweights poorly estimated pair-level effects while preserving consistency.

A fully rigorous treatment would require establishing uniform convergence of the PPML quasi-likelihood over the parameter space for each pair, extending the arguments of Newey and McFadden [1994] to the panel setting with generated regressors, and verifying the high-level conditions for pairs with highly sparse data (many zeros). We leave these technical extensions to future work, noting that our accompanying simulation evidence — which uses the exact PPML estimator across 200 replications with realistic calibration — provides strong finite-sample support for the theoretical predictions established here.

Limitations of this sketch

We acknowledge several limitations of the proof sketches presented above. First, the mapping from the linear CCE framework to the nonlinear PPML setting relies on high-level conditions (index boundedness, smoothness) rather than primitive conditions on the data-generating process; a complete treatment would derive these from structural assumptions on the model of dyadic flows. Second, the dyadic dependence structure is handled through assumptions on weak cross-pair dependence (Assumption 8) rather than through a formal characterisation of the network dependence induced by general equilibrium; the $O(1/N)$ bound on cross-pair spillovers is plausible but not derived from primitives. Third, uniform convergence of the pair-level PPML quasi-likelihood is assumed rather than established; this would require verifying conditions for each pair, including those with highly sparse data. Fourth, the symmetry assumption on υ_{ij} (Assumption 7) ensures that rreg targets the mean; relaxing this changes the estimand to the median, which may be acceptable but alters interpretation. These limitations are common to proof sketches in applied econometrics and do not, in our view, undermine the main message: the simulation evidence consistently supports the theoretical predictions across a wide range of data-generating processes.

B Further Simulation Details and Results

B.1 Data-Generating Process Details

The DGP generates bilateral flows from a structural model with N countries and T periods:

Common factors. Two AR(1) factors with persistence $\rho = 0.7$:

$$F_{k,t} = \rho \cdot F_{k,t-1} + \sqrt{1-\rho^2} \cdot \eta_{k,t}, \quad \eta_{k,t} \sim N(0,1), \quad k = 1,2 \tag{33}$$

Income. $\ln z_{1,it} = \alpha_i^{z_1} + \boldsymbol{\gamma}_i^{z_1\prime}\mathbf{F}_t + \theta_{z_1} D_{it} + u_{it}$ with $\alpha_i^{z_1} \sim N(10, 0.04)$, $\gamma_{ik} \sim N(0, 0.0225)$, $\theta_{z_1} = 0.13$, $u_{it} \sim N(0, 0.01)$.

Population. $\ln z_{2,it} = \alpha_i^{z_2} + \boldsymbol{\gamma}_i^{z_2\prime}\mathbf{F}_t + \xi_{it}$ with $\boldsymbol{\gamma}_i^{z_2} = 0.3\boldsymbol{\gamma}_i^{z_1} + \boldsymbol{\nu}_i$.

Bilateral costs. $\ln \tau_{ij,t} = \ln \bar{\tau}_{ij} + \boldsymbol{\gamma}_i^{\tau\prime}\mathbf{F}_t + \boldsymbol{\gamma}_j^{\tau\prime}\mathbf{F}_t + \frac{\theta_{\text{cost}}}{1-\sigma} D_{it}$ with $\theta_{\text{cost}} = 0.10$, $\sigma = 5$.

Treatment assignment. Which countries are treated is drawn uniformly at random (50% treated), ensuring that treated and control countries share similar loading distributions so that control CAs are valid proxies. Treatment *timing* is correlated with the global factor $F_{1,t}$:

$$t_i^* = \lfloor 0.7 \cdot t_i^{\text{factor}} + 0.3 \cdot t_i^{\text{random}} \rfloor \tag{34}$$

where t_i^{factor} is the first time $F_{1,t}$ exceeds a country-specific random threshold (drawn from the 20th–80th percentile of F_1's distribution in the treatment window $[T/4, 3T/4]$), and $t_i^{\text{random}} \sim U[T/4, 3T/4]$ provides realistic staggering. The 70% factor-driven weight creates within-pair $\text{Corr}(D_{it}, F_t) \approx 0.2$–$0.4$: countries tend to democratise during global expansions (when $F_{1,t}$ is high), generating the confounding that CAs are designed to absorb. In additional exercises we allow for correlation between treatment and income (see subsection B.3 below).

Treated factor scenario. To test robustness to spanning failures, control countries' factor loadings on the second factor are scaled by $(1-s)$:

$$\gamma_{i,2}^{z_1} \leftarrow (1-s)\,\gamma_{i,2}^{z_1}, \qquad \gamma_{i,2}^{\tau} \leftarrow (1-s)\,\gamma_{i,2}^{\tau}, \qquad i \in \text{control}$$

At $s = 0$ (baseline), all countries share the same loading distribution on both factors, so control CAs span the factor space. At $s = 1$ (extreme), control countries have zero loading on the second factor, which then loads exclusively on treated countries and cannot be proxied by control CAs. Intermediate values ($s = 0.5$) represent moderate confounding.

Equilibrium solution. Fixed-point iteration [Fally, 2015] with tolerance 10^{-10} in log space, warm-started from the previous period.

True treatment effect. Computed numerically by constructing parallel worlds with and without treatment, re-solving the equilibrium in each. $\theta^{\text{true}} \approx 0.137$ at $N = 60$, reflecting $\sim$40% GE attenuation from the partial-equilibrium prediction of 0.23.

B.2 Additional MC Results: Sample Size Scaling

Our MC results presented in the main text are based on a moderate sample of $N = 60$ exporters/importers. In Table B-1 we repeat our baseline specification (control sample CAs, $T = 65$, 50% treatment share, $s = 0$) adopting 30, 100, or 160 exporters. Note that as N rises, θ^{true} becomes smaller due to a stronger GE attenuation effect. The bias in our proposed CCE-DID specification in (1) remains small even at $N = 160$. Alternative specifications excluding CAs (2), adopting full-sample CAs instead of control sample-only CAs (3), and using MRTs (4) maintain relatively constant bias throughout this exercise.

Table B-1: Control-CA Bias Across Sample Sizes

Test	(1) Control CAs (Bias)	(2) No CAs (Bias)	(3) Dual CAs (Bias)	(4) MRT FE (Bias)	θ_{true}
$N = 30$	$+0.002$	$+0.127$	-0.066	$+0.108$	0.142
$N = 60$	$+0.004$	$+0.152$	-0.057	$+0.109$	0.137
$N = 100$	$+0.002$	$+0.156$	-0.051	$+0.104$	0.134
$N = 160$	$+0.007$	$+0.174$	-0.045	$+0.109$	0.131

Notes: Baseline specification (with control sample CAs, $T = 65$, $s = 0$, and 50% treatment share), while sample size ranges from $N = 30$ to $N = 160$ exporters. True ATET decreases with N due to GE attenuation. The control-CA bias remains negligible.

B.3 Additional MC Results: Income and Loading Structure Stratification

A natural concern in a setup where control sample unobservables are proxied and adopted to capture treated sample unobservables is that we are ‘comparing apples and oranges’. In practice, two issues may arise: first, we may encounter systemic patterns in the heterogeneous factor loadings between treated and control samples; and second, we might also worry that treatment is correlated with the level of economic development. In the following we examine whether any of these scenarios introduce notable bias into our baseline CCE-DID estimates ($N = 60$, $T = 65$, treatment share $= 50\%$, $s = 0$)

Our modified DGP introduces systematic income differences (GDP intercept shift for control countries), heterogeneous factor loadings (control loadings multiplied by $1 - \ell$), and GDP-correlated treatment selection. We consider five configurations.

- Income gap $g = 0.5$ (control countries $\exp(0.5) \approx 1.6\times$ poorer)
- Income gap $g = 1.0$ (control countries $\exp(1) \approx 2.7\times$ poorer)
- Income gap $g = 2.0$ (control countries $\exp(2) \approx 7.4\times$ poorer)
- Factor loading gap $\ell = 0.5$ (control loadings halved)
- Combined $g = 1.5$, $\ell = 0.5$ (worst case scenario)

Table B-2: Income and Factor Loading Stratification ($N = 60$, $T = 65$, 50% treated, baseline)

Test	θ^{true}	Mean $\hat{\theta}$	Bias	Std
Income gap $= 0.5$	0.129	0.125	-0.003	0.014
Income gap $= 1.0$	0.120	0.117	-0.003	0.013
Income gap $= 2.0$	0.104	0.102	-0.002	0.011
Loading gap $= 0.5$	0.137	0.129	-0.009	0.020
Combined g=1.5, ℓ=0.5	0.111	0.105	-0.006	0.015

All configurations produce small bias (-0.002 to -0.009), comparable to the baseline bias of $+0.004$. The income gap and loading gap features do not introduce meaningful additional bias. Notably, θ^{true} varies substantially across configurations (0.104 to 0.137) because the income stratification changes the GE equilibrium (poorer control countries reduce world GDP, increasing each treated country's trade share and the treatment effect). The loading gap ($\ell = 0.5$) produces the largest bias (-0.009) because reduced control loadings weaken the CA's ability to track factor-driven MRT variation — this is the mechanism tested more systematically in Table 3 of the main text where we vary spanning 'quality', s. Overall, the estimator's performance is robust to the control sample's income or loading composition.

B.4 Additional MC Results: Exclusion of Internal Trade

We investigate whether the exclusion of internal (domestic) trade, X_{ii}, from the estimation sample introduces bias in our CCE-DID estimates — considering domestic trade is strongly advocated in parts of the literature [e.g., Yotov, 2022]. Our DGP generates both international (X_{ij}, $i \neq j$) and domestic (X_{ii}) trade flows. Domestic trade is constructed as $X_{ii} = \exp(2 \ln Y_i - \ln Y^W - \ln \Phi_i - \ln \Psi_i)$ with $\tau_{ii} = 1$ (fixed, no idiosyncratic noise). Two estimation samples are compared:

1. Standard: international pairs only ($i \neq j$).
2. Expanded: all pairs including domestic trade ($i = j$).

We find that excluding domestic trade does not bias the estimator. Both specifications produce nearly identical bias ($+0.004$ in both cases) and RMSE (0.011). The estimand is

Table B-3: Internal Trade Exclusion ($N = 60$, $T = 65$, 50% treated, baseline)

Test	Estimator	Bias	RMSE
I1: International only	Control CAs	$+0.004$	0.011
I2: Intl + domestic	Control CAs	$+0.004$	0.011

well-defined on international pairs, and domestic trade inclusion is innocuous because trade diversion effects on X_{ii} average out in the MG aggregation.

B.5 Additional MC Results: Coverage and Inference

Table B-4 reports simulation results for naive one-way clustering (by exporter) and the Cameron et al. [2011] two-way clustering we suggest as favourable. The latter approach results in 94% coverage, whereas the former only manages 71%, thus confirming our suggested prefence for CGM.

Table B-4: Standard Error Coverage (95% level, $N = 60$, 200 replications)

SE Procedure	Coverage	Target
Origin-clustered, $t(G_e - 1)$	71%	95%
Two-way clustered (CGM), $t(\min(G_e, G_j) - 1)$	94%	95%

Notes: Baseline calibration ($N = 60$, $T = 65$, 50% treated, $s = 0$). Coverage = fraction of replications where the 95% CI contains the true ATET. Two-way clustering follows Cameron et al. [2011]:
$\text{Var} = \text{Var}_{\text{exporter}} + \text{Var}_{\text{importer}} - \text{Var}_{\text{pair}}$. One-way exporter clustering under-covers because it ignores importer-side general-equilibrium dependence across pairs.

C Additional Results — Democracy and Bilateral Trade

C.1 Minimum Treatment Exposure

Table C-1 compares the main ATET estimates with and without a minimum-treatment-years filter. The filter drops all pairs where the exporter has fewer than 5 years of treatment exposure, removing 10 exporters (primarily brief-switcher countries) and approximately 1,000 pairs. These short-spell pairs contribute very noisy estimates and receive near-zero weight from the rreg aggregation; their exclusion raises the ATET substantially, confirming that the trade dividend requires time to materialise.

C.2 Event Study: All Pre-Treatment Data

Table C-2 reports event-study coefficients in our baseline model (10-year pre-treatment reference window) in (1) and using all pre-treatment periods in (5). The latter are uniformly larger in magnitude due to, for some treated exporters, substantially longer pre-treatment periods.

C.3 Event Study by Destination Regime Status

Table C-2 also reports event-study coefficients (rreg aggregation) stratified by destination regime status, comparing the 10-year, columns (2) to (4), and all-pre-treatment periods reference windows, columns (6) to (8). Patterns are identical, with the estimates for all pre-treatment windows near-uniformly larger in magnitude.

Table C-1: ATET Estimates: Full Sample vs. Minimum 5 Years of Treatment Exposure

	Full sample (min=0)		Filtered (min≥5)			
	rreg	MG	rreg	MG	G	Pairs
Panel A — Main						
Benchmark	0.411***	0.440	0.514***	0.539	86→76	8,986→7,995
	(0.064)		(0.066)			
Add GDPpc	0.105***	0.103	0.146***	0.139	84→74	8,776→7,811
	(0.026)		(0.027)			
Single regime change	0.624***	0.762***	0.762***	0.851***	49→45	4,905→4,463
	(0.104)	(0.104)	(0.101)	(0.101)		
Single RC + GDPpc	0.178***	0.187***	0.231***	0.212***	48→44	4,771→4,339
	(0.041)	(0.041)	(0.045)	(0.045)		
Panel B — Robustness						
GDP-only CAs	0.283***	0.260	0.377***	0.354	86→76	8,992→7,999
	(0.060)		(0.063)			
No dyadic controls	0.449***	0.488	0.564***	0.609	86→76	9,024→8,033
	(0.068)		(0.070)			
Panel C — Proof of concept						
Add Unit Mass	−0.042	−0.000	−0.053*	−0.017	84→74	5,488→4,984
	(0.027)		(0.029)			
DepVar CA (all)	0.232***	0.253	0.304***	0.311	86→76	8,967→7,981
	(0.054)		(0.058)			
DepVar CA (control)	0.186***	0.247	0.233***	0.312	86→76	8,975→7,987
	(0.054)		(0.061)			
Add eq. terms	0.146***	0.177	0.161***	0.199	86→76	8,940→7,956
	(0.034)		(0.037)			

Notes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. The filter drops pairs where the exporter experienced fewer than 5 years of democratic treatment. Standard errors (in parentheses) use two-way CGM clustering (exporter × importer) with $t(\min(G_e, G_j)-1)$ critical values. Because taking the median in the first step eliminates most within-cluster variation the CGM standard errors for the MG aggregation are degenerate and hence not reported. The min≥5 filter removes approximately 1,000 pairs (10 exporter groups) from most specifications. Across all specifications, the filtered ATET exceeds the full-sample estimate, confirming that short-spell exporters dilute the average.

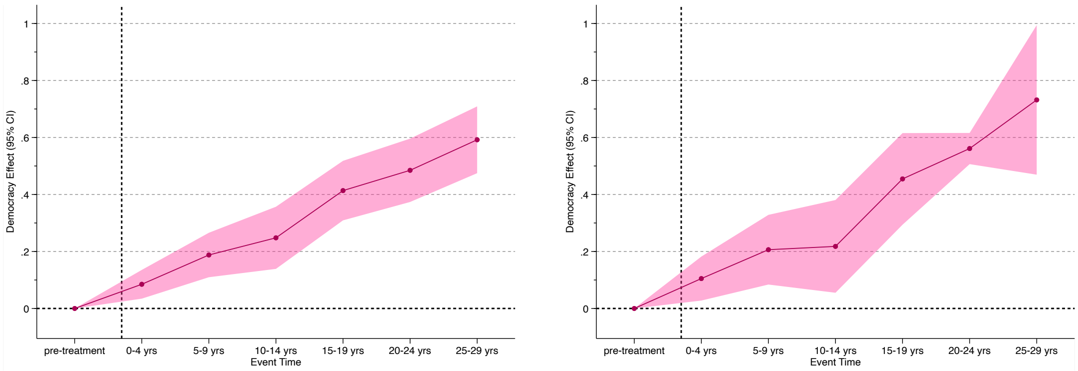
Table C-2: Event Study Estimates by Destination Regime Status

	10-year reference				All pre-treatment			
	(1) All	(2) Auto	(3) Demo	(4) Democ'r	(5) All	(6) Auto	(7) Demo	(8) Democ'r
Benchmark								
$[0, 4]$	0.065^{***}	0.030	0.099^{***}	0.071^{***}	0.085^{***}	0.047^{*}	0.138^{***}	0.082^{***}
	(0.021)	(0.025)	(0.034)	(0.025)	(0.025)	(0.027)	(0.042)	(0.028)
$[5, 9]$	0.155^{***}	0.111^{**}	0.152^{***}	0.183^{***}	0.188^{***}	0.128^{***}	0.193^{***}	0.218^{***}
	(0.031)	(0.050)	(0.051)	(0.037)	(0.039)	(0.044)	(0.062)	(0.045)
$[10, 14]$	0.223^{***}	0.158^{***}	0.283^{***}	0.236^{***}	0.248^{***}	0.184^{***}	0.304^{***}	0.258^{***}
	(0.039)	(0.059)	(0.072)	(0.042)	(0.055)	(0.066)	(0.090)	(0.060)
$[15, 19]$	0.361^{***}	0.311^{***}	0.411^{***}	0.365^{***}	0.414^{***}	0.328^{***}	0.431^{***}	0.451^{***}
	(0.044)	(0.064)	(0.057)	(0.047)	(0.052)	(0.068)	(0.075)	(0.058)
$[20, 24]$	0.405^{***}	0.304^{***}	0.539^{***}	0.403^{***}	0.485^{***}	0.341^{***}	0.594^{***}	0.512^{***}
	(0.047)	(0.059)	(0.118)	(0.053)	(0.055)	(0.063)	(0.114)	(0.063)
$[25, 29]$	0.474^{***}	0.257^{*}	0.907^{***}	0.424^{***}	0.592^{***}	0.344^{***}	1.084^{**}	0.542^{***}
	(0.031)	(0.140)	(0.265)	(0.075)	(0.057)	(0.109)	(0.471)	(0.086)
Add GDPpc								
$[0, 4]$	-0.014	0.010	-0.047	-0.001	-0.016	-0.006	-0.046^{*}	-0.005
	(0.020)	(0.031)	(0.029)	(0.024)	(0.018)	(0.025)	(0.026)	(0.020)
$[5, 9]$	0.069^{**}	0.085	-0.045	0.124^{***}	0.078^{**}	0.080^{**}	-0.036	0.140^{***}
	(0.033)	(0.061)	(0.039)	(0.037)	(0.031)	(0.040)	(0.035)	(0.036)
$[10, 14]$	0.123^{***}	0.172^{***}	-0.016	0.173^{***}	0.127^{***}	0.169^{***}	-0.005	0.180^{***}
	(0.034)	(0.051)	(0.042)	(0.041)	(0.036)	(0.051)	(0.044)	(0.040)
$[15, 19]$	0.177^{***}	0.230^{***}	-0.003	0.239^{***}	0.198^{***}	0.222^{***}	-0.003	0.286^{***}
	(0.033)	(0.057)	(0.047)	(0.041)	(0.037)	(0.056)	(0.053)	(0.044)
$[20, 24]$	0.217^{***}	0.225^{***}	0.070	0.281^{***}	0.248^{***}	0.254^{***}	0.105^{**}	0.318^{***}
	(0.036)	(0.049)	(0.046)	(0.048)	(0.042)	(0.055)	(0.044)	(0.054)
$[25, 29]$	0.129^{***}	0.061	0.021	0.214^{***}	0.178^{***}	0.132	0.046	0.273^{***}
	(0.043)	(0.075)	(0.194)	(0.053)	(0.046)	(0.081)	(0.096)	(0.056)

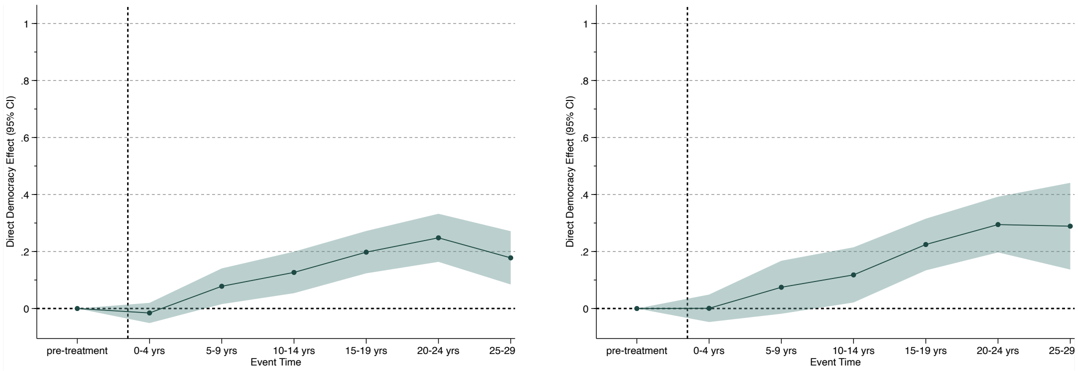
Notes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. rreg aggregation over exposed pairs only in each destination-regime subsample; pairs with zero bin-specific estimates (structurally unexposed pairs lacking sufficient treatment duration for the bin) are excluded. Auto = always-autocratic destinations; Demo = always-democratic destinations; Democ'r = democratiser destinations. Standard errors (in parentheses) use two-way clustering (exporter $\times$ importer). 10-year reference results: event window $[-10, 29]$; all pre-treatment results: no lower bound on event window. Data beyond 29 years post-treatment are too sparse for reliable robust-regression aggregation and are omitted from this event plot.

Figure C-1: Dynamic treatment effects using all pre-treatment data for the reference category

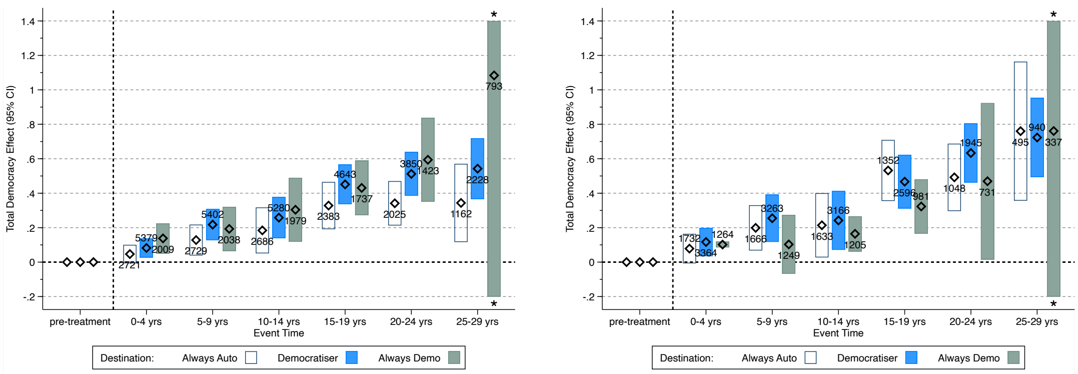
(a) Benchmark — all treated exporters (left), single regime change subsample (right)



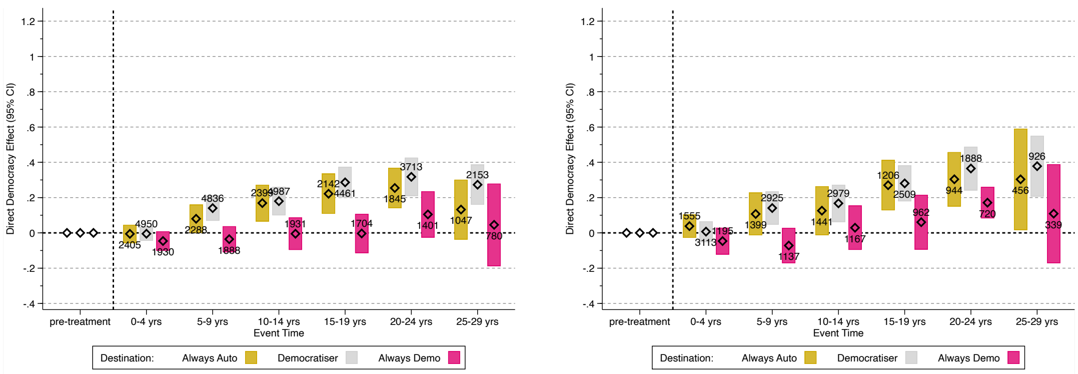
(b) Add GDPpc — all treated exporters (left), single regime change subsample (right)



(c) Benchmark by destination — all treated exporters (left), single regime change subsample (right)



(d) Add GDPpc — all treated exporters (left), single regime change subsample (right)



Notes: Same specification as Figure 2, except the event window includes all pre-treatment observations (no lower bound) rather than a fixed 10-year pre-treatment window. Post-treatment event-time bins ($[0,4]$, $[5,9]$, $[10,14]$, $[15,19]$, $[20,24]$, and $[25,29]$) relative to treatment onset; pre-treatment window normalised to zero. Shaded areas show 95% confidence intervals using two-way clustered SEs. Panels (a)–(b): all pairs, rreg aggregation; panels (c)–(d): stratified by destination regime status, pair count indicated. Left column: full sample; right column: single regime change subsample.

C.4 Carryover Test

Tables C-3 and C-4 report the full results of the carryover test described in Section 5.5. Table C-3 presents (rreg) mean-group coefficient estimates for the democracy indicator (D_{it}), the pre-democratisation dummy, and the four carry dummies (γ_1–γ_4) across destination-regime subsamples. Table C-4 reports the joint Wald test of $H_0\colon \gamma_1 = \gamma_2 = \gamma_3 = \gamma_4 = 0$, computed using 'seemingly unrelated estimates' approach described in the main text.

Table C-3: Carryover Test: Mean-Group Estimates

	(1) All	N	G	(2) Auto	N	G	(3) Demo	N	G	(4) Democ'r	N	G
Panel A — Benchmark												
D_{it}	0.668***	3381	29	0.659***	867	29	0.702***	725	29	0.650***	1789	29
	(0.095)			(0.139)			(0.142)			(0.104)		
Carry γ_1	0.369***	2621	29	0.300**	608	29	0.318**	646	29	0.429***	1367	29
	(0.096)			(0.135)			(0.139)			(0.102)		
Carry γ_2	0.472***	2403	28	0.415***	540	27	0.468***	619	28	0.502***	1244	28
	(0.108)			(0.146)			(0.156)			(0.152)		
Carry γ_3	0.504***	1643	20	0.408**	358	20	0.631***	433	20	0.468***	852	20
	(0.104)			(0.189)			(0.162)			(0.149)		
Carry γ_4	0.717***	1195	20	0.774*	227	20	0.725***	380	20	0.712***	588	20
	(0.184)			(0.468)			(0.209)			(0.209)		
Late PR	0.605***	2041	19	0.716***	504	19	0.780***	458	19	0.468***	1079	19
	(0.110)			(0.170)			(0.185)			(0.162)		
Panel B — Add GDPpc												
D_{it}	0.214***	3305	29	0.459***	829	29	0.043	723	29	0.231**	1753	29
	(0.066)			(0.121)			(0.069)			(0.101)		
Carry γ_1	0.099	2523	29	0.145	570	29	-0.075	633	29	0.221***	1320	29
	(0.066)			(0.146)			(0.066)			(0.093)		
Carry γ_2	0.153*	2340	28	0.265*	515	27	-0.013	611	28	0.240**	1214	28
	(0.088)			(0.158)			(0.081)			(0.120)		
Carry γ_3	0.156**	1596	20	0.227	333	20	0.011	426	20	0.233	837	20
	(0.072)			(0.239)			(0.093)			(0.146)		
Carry γ_4	0.247*	1220	20	0.567	236	20	-0.010	387	20	0.368*	597	20
	(0.133)			(0.446)			(0.121)			(0.207)		
Late PR	0.163**	1999	19	0.406**	483	19	0.011	453	19	0.179	1063	19
	(0.071)			(0.176)			(0.083)			(0.116)		

Notes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. rreg aggregation over reverter pairs only (29 reverting exporters). Auto = always-autocratic destinations; Demo = always-democratic destinations; Democ'r = democratiser destinations. Standard errors (in parentheses) use two-way CGM clustering (exporter $\times$ importer) with $t(\min(G_e, G_j)-1)$ critical values; G_e ranges from 19 to 29 across coefficients. N reports the count of pairwise estimates. D_{it} is the treatment dummy, Carry γ_k is for the k carryover dummies, Late PR is a dummy for all time periods in reversal *after* the first k periods.

Table C-4: Carryover Wald Test: Joint Significance of Carry Dummies

	Benchmark	pairs	+GDPpc	pairs
$\hat{\gamma}_1$	0.369	2,549	0.099	2,414
$\hat{\gamma}_2$	0.416	2,099	0.144	1,983
$\hat{\gamma}_3$	0.451	1,387	0.154	1,312
$\hat{\gamma}_4$	0.443	986	0.206	979
p-value	<0.001		0.035	
G_e (exporters)	29		29	
G_j (importers)	167		167	

Notes: $H_0: \gamma_1 = \gamma_2 = \gamma_3 = \gamma_4 = 0$. The carry coefficients $\hat{\gamma}_k$ are weighted means using a common rreg weight vector (from γ_1), ensuring consistent weighting across all four coefficients in the quadratic form. As a result, the pair count differs from that in Table C-3 above, where each Mean Group coefficient is estimated independently (i.e. weights differ between the carry coefficients). The Wald statistic uses the full 4×4 CGM variance-covariance matrix ($V_{\text{cgm}} = V_{\text{exp}} + V_{\text{imp}} - V_{\text{pair}}$). The Wald statistic for carryover is computed using seemingly unrelated estimates for robust means of each of the carryover dummies (excluding pair estimates with zero weight).

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